Topic 1 — Living World, Cell, Genetics and Biotechnology

Sources baked in: NCERT Class 11 & 12, Ghatnachakra General Science (Subdivisions of Biology, Evolution of Life, Taxonomy), Whittaker (1969), Woese (1990), UPPCS Prelims (1990–2025), UKPCS, BPSC, RAS, MPPCS, CGPCS PYQs **Weight:** ★★★ High — direct stems on biological branches, cultures, Miller-Urey gas ratio, Lamarck vs Darwin, Archaeopteryx, Cro-Magnon, cell organelles, DNA/RNA, stages of meiosis, and biotech tools **Last verified:** September 2026

Current Affairs

Year	Event / Fact	Must-Score Fact
2024	Nobel Prize in Physiology or Medicine awarded to **Victor Ambros** and **Gary Ruvkun**.	Discovered **microRNA (miRNA)** and its role in post-transcriptional gene regulation; miRNA silences mRNA after transcription.
2025	Nobel Prize in Physiology or Medicine awarded to **Mary E. Brunkow**, **Fred Ramsdell**, and **Shimon Sakaguchi**.	Discovered mechanisms of **peripheral immune tolerance** mediated by regulatory T cells (T_{reg}), preventing autoimmune attack.
2025	**GenomeIndia Project** completed by Department of Biotechnology (DBT).	Sequenced **10,000 whole genomes** across **83 distinct population groups**; housed at Indian Biological Data Centre (IBDC), Faridabad.
2025	Ministry of Agriculture approved commercial release of **two genome-edited rice varieties** developed by ICAR-IARI.	Utilise **SDN-1 / SDN-2** CRISPR edits; exempt from strict GMO transgenic rules under Rule 20 of Environment Protection Act 1989.
Current Status	**Bt cotton** remains the **only** transgenic commercial crop approved for cultivation in India (approved by GEAC in **2002**).	Bt brinjal (moratorium 2010) and DMH-11 GM mustard are not cleared for unrestricted farmer-level commercial cultivation.

1.1 Living World, Subdivisions & Terminologies of Biology

- **Science & Biological Foundations**:
- The word **'Science'** derives from the Latin word *scientia*, meaning **'knowledge'** — systematic empirical knowledge of the physical world gained through observation and experimentation.
- Science is broadly classified into: (1) **Natural Sciences** (Physical sciences, Earth and space sciences, and Life/Biological sciences) and (2) **Social Sciences**.
- **Biology** (Greek: *Bios* = life, *Logos* = study / discourse): The term was introduced independently in its modern sense in **1802** by French biologist **Jean-Baptiste Lamarck** and German naturalist **Gottfried Reinhold Treviranus**.
- **Aristotle** is celebrated as the **Father of Biology** and the **Father of Zoology**.
- **Theophrastus** is the **Father of Botany** (authored *Historia Plantarum*).
- **Antony van Leeuwenhoek** is the **Father of Microbiology**.

Master Table: Subdivisions & Specialized Branches of Biology

Branch	Exact Scope & Definition	Key Must-Score / Pioneer
Agrostology / Graminology	Scientific study of **true grasses** (family Poaceae / Gramineae)	Direct stem in Jharkhand PCS & UPPCS
Algology / Phycology	Scientific study of algae and seaweeds	Photosynthetic thallophytes
Anthology	Scientific study of **flowers**; also denotes a curated collection of literary works	Tested in Jharkhand PCS Pre 2003
Bacteriology	Study of bacteria — morphology, ecology, genetics, and biochemistry	Father: Anton van Leeuwenhoek / Robert Koch
Biometry / Biostatistics	Application of mathematical and statistical methods to analyse biological data and phenomena	Biological data analysis
Bionics	Study of mechanical systems that function like living organisms; application of biological principles to engineering and modern technology	Term coined by **Jack E. Steele (1958)** (RAS Pre 1994)
Bionomics	Comprehensive study of an organism's mode of life in its natural habitat and adaptations to its surroundings	Equivalent to **Ecology** (RAS Pre 1994)
Bionomy	Branch of biology concerned with the fundamental **laws of life**	Life laws
Botany	Detailed scientific study of plants, plant life, and physiology	Father: Theophrastus
Cell Biology / Cytology	Study of the structure, physiological properties, and function of the cell (basic unit of life)	Father: Robert Hooke
Chondrology	Specialized branch of anatomy dealing exclusively with **cartilage**	Cartilage study
Demography	Statistical and quantitative study of human populations (birth, death, migration, density)	UPPCS Pre 1994

Branch	Exact Scope & Definition	Key Must-Score / Pioneer
Dermatology	Branch of medical science concerned with the diagnosis and treatment of skin, nail, and hair disorders	Integumentary system
Ecology	Study of interactions between living organisms and their physical/biotic environment	Term coined by Ernst Haeckel (1866)
Edaphology	Study of the influence of soils on living things, particularly plant growth and crop yields	Soil-plant interaction
Embryology	Study of gametogenesis, fertilisation, and the developmental stages of embryos	Ontogeny
Entomology	Specialized branch of zoology dealing with the scientific study of **insects** (Hexapoda)	Largest class of animal kingdom
Ethics	Systematic philosophical study of moral rules and human conduct	Distinct from biological sciences
Ethnography	Descriptive scientific study of specific, individual human cultures and societies	Cultural anthropology
Ethnology	Comparative and historical study of different races, peoples, and cultures of mankind	RAS Pre 1994
Ethology	Scientific and objective study of non-human **animal behaviour** as an evolutionary adaptive trait	Konrad Lorenz, Tinbergen, von Frisch
Genetics	Scientific study of heredity, genes, and inherited variation across generations	Father: Gregor Johann Mendel (1866)
Geology	Scientific study of the Earth, its internal structure, rocks, minerals, and processes	Earth sciences
Gerontology	Multi-disciplinary study of the biological, physiological, cognitive, and social aspects of **ageing and old age**	UPPCS Pre 1993, Mains 2014
Herpetology	Branch of zoology dealing with **amphibians and reptiles** (frogs, lizards, snakes, turtles, crocodilians)	Ectothermic tetrapods
Histology	Microscopic study of the anatomical structure of animal and plant tissues	Father: Marie François Xavier Bichat
Ichthyology	Branch of zoology devoted to the scientific study of **fishes** (Pisces)	Fish science
Iconography	Study of visual images, symbols, portraits, and traditional idols in art history	Cultural art history
Lexicography	The art, craft, and science of **compiling, writing, and editing dictionaries**	Chhattisgarh PCS Pre 2008
Lichenology	Scientific study of lichens (symbiotic mutualism of fungi and algae/cyanobacteria)	SO_2 pollution bioindicators
Malacology	Branch of invertebrate zoology dealing with **molluscs** (snails, clams, octopuses, squids) and their shells	Second-largest animal phylum
Mycology	Scientific study of **fungi** (yeasts, moulds, mushrooms)	Heterotrophic chitinous organisms
Oncology	Specialized branch of medicine dealing with the prevention, diagnosis, and treatment of **tumours and cancer**	Oncogenes, biopsy, radiation
Ornithology	Branch of zoology dedicated to the scientific study of **birds** (Aves)	Bird science (Salim Ali: Birdman of India)
Orology	Scientific study of **mountains** and their topographic formation	Physical geography
Osteology	Scientific and detailed study of **bones**, skeletal elements, teeth, and bony morphology	UPPCS Pre 1990, 65th BPSC Pre 2019
Palaeobotany	Study of **plant fossils** and past vegetative environments	Father of Indian Palaeobotany: **Prof. Birbal Sahni**
Palaeontology	Study of the history of life on Earth based on **fossils** preserved in geological strata	Father: Leonardo da Vinci / Georges Cuvier
Palynology	Study of **pollen grains and plant spores** (used for radiocarbon dating and reconstructing past climates)	Paleoclimate & fossil pollen
Parasitology	Study of parasites (protozoa, helminths, ectoparasites), their host organisms, and host-parasite dynamics	Host-parasite biology
Pathology	Study of the nature, causes, and structural/functional effects of diseases via laboratory tissue examination	Diagnostic medicine
Pedology	Scientific study of **soils in their natural environment**, pedogenesis, soil morphology, and classification	RAS Pre 1999
Philology	Study of language structure, historical development, and literary linguistics	Historical linguistics
Phrenology	Study of the shape, bumps, and size of the human **cranium (skull)** as an indicator of character/mental faculties	Skull anatomy
Physiology	Study of normal mechanical, physical, and biochemical functions of living organisms and their organs	Normal body function
Psychology	Scientific study of the human mind, cognition, and behavioural processes	Distinct from pure biological branches
Serology	Diagnostic examination and study of **blood serum**, especially antigen-antibody immune responses	Serum & antibodies
Serpentology / Ophiology	Specialized branch of zoology dealing with the study of **snakes**	Snake biology
Teratology	Scientific study of **congenital abnormalities, birth defects, and physiological malformations**	Developmental abnormalities

Branch	Exact Scope & Definition	Key Must-Score / Pioneer
Thanatology	Scientific study of **death**, dying processes, and associated social and psychological mechanisms	UPPCS Mains 2014
Trophology	Specialized branch of science dealing with **nutrition** and dietary requirements	Nutritional science
Virology	Study of viruses — ultramicroscopic parasitic nucleoprotein particles enclosed in protein capsids	Capsid & genetic material
Zoology	Detailed scientific study of animals, animal kingdom members, and animal life in general	Father: Aristotle (deals with living and dead)

Master Table: Agricultural, Biological & Culture Terminologies

Terminology / Culture	Target Organism / Commodity	Practical Scope, Key Products & Key Highlights
Aeroculture	Soilless plants	Growing plants suspended above continuous liquid sprays that moisten roots with nutrient water
Aeroponics	Soilless plants	High-tech plant cultivation without soil; roots hang suspended in air misted periodically with nutrient solution
Algaculture	Algae / Microalgae	Cultivation of algae for food, biofuel, wastewater bioremediation, nutrient recovery, and biofertilisers
Agriculture	Crops & Livestock	Practice of cultivating soil, growing edible/cash crops, and rearing animals for food, wool, and products
Agronomy	Field Crops & Soil	Specialized branch of agriculture dealing with field crop production, soil management, and tillage
Apiculture	Honeybees (*Apis*)	Rearing and management of bees for natural honey, beeswax, royal jelly, propolis, and crop pollination services
Aquaculture	Aquatic organisms	Commercial farming of freshwater and marine fish, crustaceans, molluscs, and aquatic plants
Arboriculture	Trees & Shrubs	Cultivation, management, and individual study of perennial woody plants, trees, and ornamental shrubs
Aviculture	Birds	Breeding, rearing, and keeping of wild and domestic birds in captivity
Citriculture	Citrus fruits	Cultivation and commercial orchard management of citrus trees (oranges, lemons, limes, grapefruits)
Floriculture	Flowering plants	Discipline of horticulture concerned with cultivating flowering and ornamental foliage plants for gardens/floristry
Heliculture	Land snails	Snail farming to harvest edible snail meat (escargot), cosmetic snail slime, or snail caviar
Horticulture	Garden crops	Broad science and art of cultivating fruits, vegetables, flowers, spices, condiments, and plantation crops
Hydroponics	Soilless plants	Cultivation of crops in nutrient-enriched water solutions without soil; uses inert media like perlite, sand, or gravel
Mariculture	Marine organisms	Specialized aquaculture branch culturing marine organisms (shrimp, oysters, seaweeds, fish) in open seawater
Moriculture	Mulberry (*Morus alba*)	Cultivation of mulberry plants exclusively to provide fresh leaves for rearing silkworm caterpillars (*Bombyx mori*)
Olericulture	Vegetables	Science of vegetable growing, dealing strictly with non-woody (herbaceous) food plants
Oyster Culture / Pearl Culture	Oysters (*Pinctada*)	Breeding and farming oysters for food; commercial insertion of nuclei into pearl oysters to induce nacre/pearl formation
Pisciculture	Fishes	Controlled breeding, rearing, and hatching of fishes in artificial ponds, tanks, and hatcheries
Pomology	Fruits	Specialized branch of horticulture dealing with fruit culture, orchard production, and pomological science
Sericulture	Silkworms (*Bombyx*)	Rearing of silkworm caterpillars for commercial production of natural raw silk fibres from cocoons
Silviculture	Forest trees	Theory and practice of controlling the establishment, growth, composition, health, and quality of forest stands
Spongiculture	Marine sponges	Cultivation and commercial harvesting of marine bath sponges
Vegeculture	Root & Tuber crops	Agricultural systems producing asexually / vegetatively propagated root and tuber crops (cassava, yam, taro)
Vermiculture	Earthworms	Management and cultivation of earthworms to digest organic wastes into nutrient-rich vermicompost fertilizer
Viticulture	Grapevines (*Vitis*)	Science, commercial cultivation, and harvesting of grapes for table fruit, raisins, and wine production

- **Defining vs Non-Defining Features of Life**:
- **Metabolism**: Sum total of all anabolic (constructive) and catabolic (destructive) reactions occurring in the body. **Defining feature** (no non-living object exhibits metabolism).
- **Cellular Organisation**: Every living being consists of one or more cells. **Defining feature**.
- **Consciousness**: Ability to sense the environmental surroundings and respond to external physical, chemical, or biological stimuli. **Defining feature**.

- **Growth & Reproduction**: **Not** absolute defining features because non-living objects (mountains, sand dunes) grow by extrinsic surface accumulation, while several living organisms (mules, sterile worker bees, infertile human couples) never reproduce.
- **Acellular Infectious Entities (Outside the Kingdoms)**:
- **Viruses**: Ultramicroscopic, obligate intracellular parasites consisting of a genetic core (either DNA or RNA, **never both**) enclosed in a protein capsid (composed of capsomeres). First discovered by **D. I. Ivanovsky (1892)** in Tobacco Mosaic Disease; crystallized by **W. M. Stanley (1935)**; Beijerinck called fluid *Contagium vivum fluidum*.
- **Viroids**: Discovered by **T. O. Diener (1971)**. Infectious free RNA molecules devoid of any protein coat; molecular weight is lower than viral RNA. Causes **Potato Spindle Tuber disease**.
- **Prions**: Proteinaceous infectious particles devoid of nucleic acid; highly resistant to heat, radiation, and enzymes. Cause **Bovine Spongiform Encephalopathy (BSE / Mad Cow Disease)** in cattle and **Creutzfeldt-Jakob Disease (CJD)** in humans.
- **Lichens**: Symbiotic association between an alga (**Phycobiont**, autotrophic) and a fungus (**Mycobiont**, heterotrophic); excellent bio-indicators of SO_2 **air pollution** (do not grow in polluted industrial areas).

> **Note:** UPPCS 2021 tested: Nitrogen is an essential constituent of DNA, RNA, and chlorophyll, but is **absent** in simple carbohydrates.

1.2 Origin of Life & Prebiotic Chemical Evolution

- **Cosmic & Planetary Differentiation Phase**:
- Earth formed approximately **4.54 billion years ago** by accretion from the rotating solar nebula (~one-third of the Universe's current estimated age of 13.8 billion years).
- During Earth's initial molten differentiation, dense metals (chiefly **Iron and Nickel**) sank toward the core, while silicates and lighter elements formed the mantle and crust.
- Lighter atoms (Hydrogen, Helium, Nitrogen, Oxygen, Carbon) escaped or formed the primordial atmosphere.
- **The Primitive Atmosphere — Strongly Reducing (NO Free Oxygen)**:
- The primitive atmosphere was **strictly reducing**; it contained **Methane (CH_4), Ammonia (NH_3), Hydrogen (H_2), Water Vapour (H_2O), Carbon Dioxide (CO_2), Nitrogen (N_2), Carbon Monoxide (CO), and Hydrogen Cyanide (HCN)**.
- **Absolute High-Yield Fact: Free molecular gaseous Oxygen ($\mathbf{O_2}$) was completely ABSENT** on primitive Earth!
- There was **no protective ozone (O_3) shield**; intense ultraviolet (UV) radiation, cosmic rays, volcanic heat, and lightning discharges directly irradiated the prebiotic Earth.
- **The Oparin-Haldane Hypothesis (Chemogeny / Prebiotic Evolution)**:
- Formulated independently by Russian biochemist **Aleksandr I. Oparin (1924, book *Origin of Life* 1936)** and British biologist **J. B. S. Haldane (1929)**.
- Postulates that life arose on early Earth spontaneously from **non-living inorganic matter through progressive chemical evolution (Abiogenesis)**.
- As ancient terrestrial rains cooled the crust, dissolved minerals, gases, and newly synthesised simple organic molecules (amino acids, fatty acids, purines, pyrimidines) accumulated in the oceans.
- Haldane famously coined the term **"Hot Dilute Soup"** (or **Primordial Soup**) for this oceanic chemical broth, where organic molecules accumulated without decaying because bacterial life and oxygen were absent.
- Organic macromolecules aggregated into primitive non-living microscopic clusters called **Coacervates** (Oparin) or **Microspheres** (Sidney Fox), which possessed semi-permeable boundaries, absorbed chemicals, and grew before dividing.
- **The Stanley Miller & Harold Urey Landmark Experiment (1953)**:
- Conducted at the University of Chicago to experimentally test whether organic molecules could synthesize spontaneously under simulated primitive Earth conditions.
- **Experimental Design**:
  - Built a closed, airtight, sterilised borosilicate glass apparatus.
  - Circulated four continuous gases: **Methane ($\mathbf{CH_4}$), Ammonia ($\mathbf{NH_3}$), Hydrogen ($\mathbf{H_2}$)** along with boiling water vapour ($\mathbf{H_2O}$).
  - **Exact Gaseous Ratio: $\mathbf{CH_4 : NH_3 : H_2 = 2 : 2 : 1}$**.
  - Passed continuous electric spark discharges (using tungsten electrodes at **60,000 volts** simulating lightning) at ~$\sim$ **800°C**.
  - Circulated the gaseous mixture continuously for **one full week**, then condensed and cooled the fluid.
- **Landmark Result**:
  - Chromatographic analysis revealed numerous simple organic compounds, prominently amino acids: **Glycine, Alanine, and Aspartic acid**, proving that the molecular building blocks of proteins and life could form abiogenetically.
- **Chronology of Early Life**:
- Earliest fossil evidence indicates primitive microbial prokaryotic life existed by $\sim$ **3.5 to 3.8 billion years ago** (e.g., stromatolites).
- **Cyanobacteria (Blue-Green Algae)** evolved oxygenic photosynthesis, releasing free oxygen that reacted with dissolved iron and gradually converted the reducing atmosphere into an **oxidizing atmosphere** (the Great Oxidation Event).
- **Eukaryotic cells** with true membrane-bound nuclei evolved approximately **1.7 to 1.9 billion years ago**.
- **Geological Chronology of Cenozoic Epochs (Oldest to Most Recent)**:

$$\textbf{Paleocene} \rightarrow \textbf{Eocene} \rightarrow \textbf{Oligocene} \rightarrow \textbf{Miocene} \rightarrow \textbf{Pliocene} \rightarrow \textbf{Pleistocene} \rightarrow \textbf{Holocene (Current)}$$

(Must-Score: Tested in Chhattisgarh PCS Pre 2021 — Eocene precedes Miocene, Pliocene, Pleistocene, and Holocene).

- **Earth's Magnetic Field Reversals**:

- Earth's geomagnetic field has inverted polarity periodically every few hundred thousand years (over 20 times in the last 5 million years; the last major reversal was the Brunhes-Matuyama reversal ~780,000 years ago).

1.3 Theories of Organic Evolution & Landmark Evidence

- **Concept of Organic Evolution**:
- Defined as *"descent with modification"* — the process by which present-day complex plants and animals evolved gradually over millions of years through structural modifications from simpler, ancestral organisms.
- Organic evolution explains **diversification and adaptive change** after life appeared, rather than the initial origin of the first living cell.

1. Lamarckism (Theory of Inheritance of Acquired Characteristics)

- Proposed by French naturalist **Jean-Baptiste de Lamarck (1744–1829)**; first broad, comprehensive theory of organic evolution.
- Expounded in his seminal work ***Philosophie Zoologique* (1809)** (first outlined in 1801–1802).
- **Core Principles**: 1. **Internal Vital Force / Tendency to Increasing Complexity**: Organisms naturally tend to increase in size and complexity. 2. **Direct Environmental Influence & New Needs**: Altered environments create new needs (*besoins*), prompting new habits and actions. 3. **Use and Disuse of Organs**: Continuous use of an organ strengthens, enlarges, and develops it; continuous disuse causes gradual atrophy, degeneration, and eventual disappearance. 4. **Inheritance of Acquired Characters**: Structural changes acquired by an individual during its lifetime in response to the environment are directly transmitted to its offspring.
- **Classic Lamarckian Examples**:
- **Giraffe's Neck**: Ancestral short-necked giraffes stretched their necks and forelimbs to browse leaves on tall acacia trees; repeated stretching over successive generations caused gradual elongation of neck and forelimbs.
- **Loss of Limbs in Snakes**: Ancestral snakes possessed limbs; adapting to burrowing in narrow holes and creeping through dense vegetation, limbs were disused and gradually lost.
- **Webbed Feet of Aquatic Birds**: Ducks and geese spread their toes while swimming; sustained skin stretching between digits produced interdigital webs.
- **Long Legs of Wading Birds**: Sustained stretching to keep bodies above water level produced elongated legs.
- **Whales**: Lost their functional hindlimbs due to aquatic lifestyle.

2. Darwinism (Theory of Natural Selection)

- Propounded by English naturalist **Charles Robert Darwin (1809–1882)**.
- Formulated after his 5-year voyage on the survey ship **HMS Beagle (1831–1836)**, notably observing finches and giant tortoises on the **Galápagos Islands**.
- Joint paper with **Alfred Russel Wallace** presented in 1858 at the Linnean Society of London.
- Published his masterwork: ***On the Origin of Species by Means of Natural Selection, or the Preservation of Favoured Races in the Struggle for Life* (November 24, 1859)**.
- **Five Core Pillars of Darwinian Natural Selection**: 1. **Enormous Overproduction of Offspring (Prodigality of Nature)**: All living species produce far more progeny than can possibly survive (e.g., an oyster lays millions of eggs per spawn). 2. **Limited Natural Resources & Constant Population**: Food, water, and space remain relatively constant, preventing exponential growth. 3. **Struggle for Existence**: Severe competition across three fronts:
  - *Intraspecific struggle*: Between members of the same species for identical food and mates (fiercest struggle).
  - *Interspecific struggle*: Between different species (predator vs prey).
  - *Environmental struggle*: Against adverse physical forces (droughts, floods, volcanic eruptions, tsunamis, earthquakes). 4. **Universal Variations**: No two individuals in a sexually reproducing population are identical; individuals vary in morphological and physiological traits. 5. **Natural Selection ("Survival of the Fittest")**: Individuals possessing favorable heritable variations that aid survival reproduce more successfully, leaving more offspring (**differential reproduction**). Unfavorable variations are progressively eliminated.
- *Must-Score:* The famous phrase **"Survival of the Fittest"** was coined by philosopher **Herbert Spencer** in his *Principles of Biology* (1864) and adopted by Darwin in his 5th edition (1869).

3. Neo-Darwinism & The Modern Synthetic Theory

- **August Weismann's Germplasm Theory (1892)**: Decisively disproved Lamarckian inheritance of acquired characters by cutting off the tails of mice for 22 successive generations; offspring were always born with normal long tails. Proved that only changes in the **germplasm** (germ cells / gametes) are heritable, whereas changes in **somatoplasm** (body somatic tissues) perish with the individual.
- **Neo-Darwinism**: Term coined by **George Romanes in 1895** to denote Darwin's theory purged of Lamarckian acquired inheritance and reconciled with early Mendelian genetics.
- **The Modern Synthetic Theory of Evolution**:
- The comprehensive prevailing evolutionary framework synthesising Darwinian natural selection with Mendelian genetics, population genetics, and systematics.
- Formulated by **Theodosius Dobzhansky** (*Genetics and the Origin of Species*, 1937), **R. A. Fisher, J. B. S. Haldane, Sewall Wright, Ernst Mayr** (*Systematics and the Origin of Species*, 1942), and **G. L. Stebbins** (*Processes of Organic Evolution*, 1966).
- Identifies **5 basic evolutionary factors**:
  1. **Gene Mutations**: Permanent alterations in nucleotide sequences (ultimate source of new alleles).
  2. **Chromosomal Aberrations & Recombination**: Shuffling of genes during crossing over and independent assortment.
  3. **Gene Flow & Genetic Drift**: Allelic migration between populations and random fluctuations in allele frequencies in small populations (Sewall Wright effect).

4. **Natural Selection**: Differential reproductive success of genotypes.
5. **Reproductive Isolation**: Geographic or biological barriers preventing interbreeding between divergent populations, consolidating **speciation**.

4. Hugo de Vries' Mutation Theory

- Proposed by Dutch botanist **Hugo de Vries** in his treatise ***The Mutation Theory*** **(Vol. I 1901, Vol. II 1903)**.
- Derived from continuous breeding observations on the **Evening Primrose** (***Oenothera lamarckiana***).
- **Core Principles**:
- Evolution is not a slow, continuous gradual process (as envisioned by Darwin), but a **discontinuous, jerky, step-wise process**.
- New species arise suddenly due to large, single-step spontaneous inheritable changes termed **Mutations** (**Saltation** = single-step large mutation).
- Mutations are random, directionless, and operational immediately; only germline mutations are transmitted to progeny.
- Mutations serve as the **ultimate raw material** for evolutionary change and diversity.

Historical Theory / Concept	Pioneer Scientist(s)	Landmark Work / Organism Studied
Theory of Mutation	**Hugo de Vries** (1901)	Evening Primrose (*Oenothera lamarckiana*)
Theory of Natural Selection	**Charles Darwin** (1859)	*On the Origin of Species*; Galápagos Finches
Inheritance of Acquired Traits	**Jean-Baptiste Lamarck** (1809)	*Philosophie Zoologique*; Giraffes, Snakes
One Gene — One Enzyme Hypothesis	**George Beadle & Edward Tatum** (1941)	Red bread mould (*Neurospora crassa*)
Operon Concept of Gene Regulation	**François Jacob & Jacques Monod** (1961)	*Lac Operon* in *Escherichia coli*
Germplasm Theory	**August Weismann** (1892)	Mouse tail amputation experiments
Coacervate Origin of Life	**Aleksandr I. Oparin** (1936)	*The Origin of Life* (Primordial Soup)
Prebiotic Spark Discharge Synthesis	**Stanley Miller & Harold Urey** (1953)	CH_4, NH_3, H_2, H_2O (Ratio $2:2:1$)

Landmark Missing Links, Evolutionary Fossils & Human Ancestry

- ***Archaeopteryx lithographica***:
- Discovered in **1861** in the Upper Jurassic lithographic limestone quarries of **Solnhofen, Bavaria, Germany** (~150 million years old).
- Universally recognised as the premier **connecting link between Reptilia and Aves (Reptiles and Birds)**, providing undeniable evidence that birds evolved directly from theropod dinosaurian reptiles.
- **Reptilian Characteristics**: Jaws equipped with sharp homodont teeth; long lizard-like bony tail with 20 vertebrae; wings bearing three clawed digits; solid non-pneumatic bones; abdominal ribs (gastralia).
- **Avian Characteristics**: Body covered with aerodynamic asymmetrical contour feathers; forelimbs modified into wings; fused wishbone (**furcula**); large bird-like eyes and rounded cranium.
- **Dinosaurs & The Mesozoic Era**:
- The **Mesozoic Era** (251.9 to 66 million years ago) is universally celebrated as the **"Age of Reptiles"** or **"Age of Dinosaurs"**.
- Divided into three geological periods:
  1. **Triassic Period** (251.9 to 201.4 mya): First dinosaurs and primitive mammals appear.
  2. **Jurassic Period** (201.4 to 143.1 mya): Peak of giant sauropod dinosaurs; *Archaeopteryx* appears.
  3. **Cretaceous Period** (143.1 to 66 mya): Climax and catastrophic mass extinction of non-avian dinosaurs (K-Pg extinction asteroid event).
- **Woolly Mammoth**:
- Extinct hairy proboscidean species of the genus *Mammuthus*; lived during the Pliocene and Pleistocene epochs; ancestral relative sharing a common ancestor with modern Asian and African **elephants** (~6 million years ago).
- **Denisovans (Denisova Hominins)**:
- Extinct archaic human species/subspecies that ranged across Asia during the Lower and Middle Paleolithic.
- First identified in **2010** via mitochondrial DNA (mtDNA) extracted from a juvenile female finger bone discovered in **2008 in Denisova Cave (Altai Mountains, Siberia)**; additional specimens identified in **Baishiya Karst Cave on the Tibetan Plateau**.
- Nuclear DNA confirms close evolutionary affinities with Neanderthals; possessed exceptionally large molars and Denisovan genes (e.g. *EPAS1* for high-altitude hypoxia adaptation in Tibetans).
- **Cro-Magnon Man (Early Modern *Homo sapiens*)**:
- First discovered in **1868** by French palaeontologist **Louis Lartet** in the Cro-Magnon rock shelter at Les Eyzies, Dordogne, southwestern France.
- Remains radiocarbon dated to **30,000–45,000 years ago**; regarded as the **last direct ancestor of modern humans**.
- **Anatomical Facts**: Robust muscular build (height 1.7–1.8 m); high, straight vertical forehead; short, wide face with a **prominent modern chin**; large cranial capacity averaging $\mathbf{1,600\ cc}$ (slightly larger than modern average humans ~1,350 cc).
- **Vertebrate Evolutionary Sequence**:

$$\textbf{Salamander (tailed amphibian, } \sim \textbf{360 mya)} \rightarrow \textbf{Python (squamate reptile, } \sim \textbf{300 mya)} \rightarrow \textbf{Kangaroo (marsupial mammal, } \sim \textbf{145 my}$$

(Must-Score: Tested in IAS Pre 2009).

1.4 Taxonomy, Nomenclature & The Five Kingdoms

- **Concept & Pioneers of Taxonomy**:
- **Taxonomy** is the science and practice of identification, nomenclature, and classification of living and extinct organisms into hierarchical groups.
- **Augustin Pyramus de Candolle (A. P. de Candolle)**: Swiss botanist who **first coined the term 'Taxonomy'** in 1813 (from Greek *taxis* = arrangement and *nomos* = law) and was the **first to use the characteristics of vascular tissues in the classification of plants**, categorising plants into *Vasculares* (with vascular bundles) and *Cellulares* (without vascular bundles). *(Must-Score: 71st BPSC Pre 2025)*.
- **Carolus Linnaeus** (1707–1778): Swedish physician and botanist widely revered as the **Father of Modern Taxonomy**; formalised **Binomial Nomenclature** and established the Linnaean taxonomic hierarchy.
- **Binomial Nomenclature (Linnaeus System)**:
- Introduced by **Carolus Linnaeus** in *Systema Naturae* (1758) and *Species Plantarum* (1753).
- Each organism has two Latinised components: **Generic name (Genus)** + **Specific epithet (Species)**.
- Rules: Genus begins with a capital letter; species begins with a lowercase letter; both are printed in *italics* or underlined separately when handwritten (e.g., *Homo sapiens*, *Mangifera indica*).
- **Taxonomic Hierarchy (Linnaean Hierarchy)**:
- Obligate categories from broadest to most specific: **Kingdom → Phylum (Animals) / Division (Plants) → Class → Order → Family → Genus → Species**.
- In descending hierarchy: **Plant Kingdom → Phylum / Division → Class → Order → Family → Genus → Species** *(Must-Score: UPPCS Pre 2024, Uttarakhand PCS Pre 2025)*.
- As we move from Species to Kingdom, the number of shared common characteristics **decreases**, while the difficulty of determining phylogenetic relationship increases.
- **Species** is the lowest, basic, real and fundamental unit of classification: defined as a group of individual organisms with fundamental similarities capable of naturally interbreeding and producing fertile offspring *(Chhattisgarh PCS Pre 2024)*.
- **Evolution of Classification Systems**:
- **Two-Kingdom System**: Linnaeus (Plantae, Animalia) — failed to distinguish prokaryotes from eukaryotes, unicellular from multicellular, photosynthetic from non-photosynthetic.
- **Three-Kingdom System**: Ernst Haeckel (1866) — added Kingdom **Protista** for unicellular organisms.
- **Four-Kingdom System**: Herbert F. Copeland (1956) — added Kingdom **Monera** for prokaryotes.
- **Five-Kingdom System**: **R. H. Whittaker (1969)** — criteria: Cell structure, Thallus organisation, Mode of nutrition, Reproduction, and Phylogenetic relationships.
- **Whittaker's Five Kingdoms at a Glance**:

Kingdom	Cell Type	Nuclear Envelope	Cell Wall	Mode of Nutrition	Key Examples
Monera	Prokaryotic	Absent	Non-cellulosic (Peptidoglycan / Murein)	Autotrophic (phototrophic/chemotrophic) or Heterotrophic (saprophytic/parasitic)	Bacteria, Cyanobacteria (BGA), Archaebacteria, *Mycoplasma*
Protista	Eukaryotic	Present	Present in some (cellulose in Dinoflagellates)	Autotrophic (photosynthetic) or Heterotrophic	*Amoeba*, *Paramecium*, *Euglena*, Diatoms, Dinoflagellates, Slime moulds
Fungi	Eukaryotic	Present	Present (**Chitin** / fungal cellulose)	Heterotrophic (strictly saprophytic or parasitic; absorb dissolved organic matter)	Yeast (*Saccharomyces*), *Penicillium*, *Agaricus* (mushroom), *Rhizopus*
Plantae	Eukaryotic	Present	Present (**Cellulose** + hemicellulose)	Autotrophic (photosynthetic)	Algae, Bryophytes, Pteridophytes, Gymnosperms, Angiosperms
Animalia	Eukaryotic	Present	**Absent**	Heterotrophic (Holozoic / ingestion, saprozoic)	Sponges, Worms, Insects, Fish, Amphibians, Reptiles, Birds, Mammals

- **Three-Domain System (Carl Woese, 1990)**:
- Divided Kingdom Monera into two separate domains based on **16S ribosomal RNA (rRNA)** gene sequences:
  1. **Domain Archaea**: Ancient prokaryotes with branched ether-linked cell membrane lipids lacking peptidoglycan. Subdivided into **Methanogens** (anaerobic, produce methane in marshes and ruminant guts), **Halophiles** (survive extreme hypersaline conditions like the Great Salt Lake), and **Thermoacidophiles** (thrive at $80°C$ and pH 2 in sulfur springs).
  2. **Domain Bacteria**: True bacteria (Eubacteria) having ester-linked lipids and peptidoglycan cell walls.
  3. **Domain Eukarya**: All eukaryotic organisms (Protista, Fungi, Plantae, Animalia).

1.5 Cell Structure: Prokaryote vs Eukaryote & Cell Theory

- **Ghatnachakra High-Yield Facts & Cell Facts**:
- **Total Cell Count**: A standard adult human of 70 kg contains approximately **30 trillion (3×10^{13})** cells.
- **Cell Dimensions**: Most plant and animal cells measure between **1 to 100 micrometres (μm)** and are visible only under a light/electron microscope.
- **Evolutionary Emergence**: The first cellular life emerged on Earth approximately **3.5 to 3.8 billion years ago** in the form of anaerobic, heterotrophic prokaryotes.
- **Ghatnachakra High-Yield Facts & Cell Facts**:
- **Total Cell Count**: A standard adult human of 70 kg contains approximately **30 trillion (3×10^{13})** cells.

- **Cell Dimensions**: Most plant and animal cells measure between **1 to 100 micrometres (μm)** and are visible only under a light/electron microscope.
- **Evolutionary Emergence**: The first cellular life emerged on Earth approximately **3.5 to 3.8 billion years ago** in the form of anaerobic, heterotrophic prokaryotes.
- **Milestones in Cell Discovery**:
- **1665 — Robert Hooke**: Examined thin slices of bottle cork under a crude compound microscope; observed box-like porous compartments and coined the term "Cell" (Latin *cellula* = small room) in his book *Micrographia*. (Observed dead cell walls only).
- **1674 — Anton van Leeuwenhoek**: First observed and described free **living cells** ("animalcules"), including bacteria, spirogyra, protozoa, and human red blood cells.
- **1831 — Robert Brown**: Discovered and named the **Nucleus** within orchid root cells.
- **1839 — J. E. Purkinje**: Coined the term **Protoplasm** for the living fluid substance of the cell; Thomas Huxley famously termed protoplasm the *"physical basis of life"*.
- **The Cell Theory**:
- Formulated jointly by German botanist **Matthias Schleiden (1838)** and German zoologist **Theodor Schwann (1839)**:
    1. All living organisms are composed of cells and products of cells.
    2. The cell is the structural and functional unit of all living beings.
- **Modification by Rudolf Virchow (1855)**:
    - Added the foundational principle ***Omnis cellula-e cellula***: New cells arise only through the division of pre-existing cells.
- **Exceptions to Cell Theory**:
    - **Viruses**: Sub-cellular, acellular particles devoid of cytoplasm, organelles, and metabolic machinery.
    - Viroids and Prions.
    - Coenocytic / Syncytial organisms (multinucleated single masses, e.g. *Rhizopus*, *Vaucheria*, skeletal muscle fibres).
- **Prokaryotic Cell vs Eukaryotic Cell**:

Feature	Prokaryotic Cell (e.g. Bacteria, BGA)	Eukaryotic Cell (e.g. Plant, Animal, Fungi)
Size	Typically small (1–10 μm)	Larger (10–100 μm)
Nuclear Structure	**Nucleoid** (no nuclear membrane, no nucleolus)	**True Nucleus** (double membrane, nucleolus present)
Genetic Material	Single, circular double-stranded DNA; **no histones**	Multiple linear double-stranded DNA complexed with **histone proteins**
Ribosomes	**70S** (50S large subunit + 30S small subunit)	**80S** in cytoplasm (60S + 40S); 70S inside mitochondria & chloroplasts
Membrane Organelles	**Absent** (no ER, Golgi, lysosomes, mitochondria, plastids)	**Present** (ER, Golgi, mitochondria, lysosomes, plastids)
Respiratory Site	**Mesosomes** (infoldings of plasma membrane)	**Mitochondria** inner membrane (cristae)
Cell Wall	Complex peptidoglycan (murein)	Cellulose (plants), Chitin (fungi), Absent (animals)
Cell Division	Binary fission, budding (amitosis; no spindle formation)	Mitosis and Meiosis (spindle apparatus forms)

> **Note:** UPPCS 2024 tested: Chromosomes are made of DNA and protein (histones); the nucleus **does** play an active role in cellular reproduction.

1.6 Detailed Organelles & Cellular Machinery

- **Ghatnachakra High-Yield Organelle Facts**:
- **Sarcoplasmic Reticulum (SR)**: A modified smooth endoplasmic reticulum (SER) present *exclusively in muscle cells (myocytes)*; functions to sequester, store, and pump calcium ions (Ca^{2+}) for muscle contraction and relaxation.
- **Mitochondrial ATP Respiration Breakdown**: While cellular respiration theoretically generates 30–38 ATP per glucose molecule, the **terminal oxidation process / oxidative phosphorylation** (electron transport chain linked to ATP synthase in inner mitochondrial membrane) produces the vast majority—**approx. 27–34 ATP**!
- **Golgi Apparatus**: First observed in 1897 and officially reported in April 1898 by Italian physician **Camillo Golgi**; composed of stacks of membrane-bound flattened sacs called **cisternae** responsible for modifying, sorting, and packaging proteins and lipids.
- **Lysosomes ("Suicide Bags")**: Discovered by Christian de Duve; vesicular organelles packed with acid hydrolases (proteases, lipases, nucleases, carbohydrases) that function optimally in an **acidic medium (pH ~4.5–5.0)**. Functions in autophagy (digesting obsolete organelles), autolysis (cell self-destruction upon rupture), and bactericidal killing.
- **Centrosome**: The primary microtubule organizing centre (MTOC) of animal cells; contains two centrioles surrounded by pericentriolar material; organizes the spindle apparatus during mitosis. **Must-Score Fact:** Plants and most fungi lack centrosomes!
- **Lysosome vs Vacuole Taxonomic Convention**: In scientific convention, the term *lysosome* is strictly applied to animal vesicular digestive organelles, whereas in plants, fungi, and algae, these functions are performed by the large central **vacuole**.
- **Programmed Cell Death (PCD) / Apoptosis ("Cellular Suicide")**: Regulated intracellular suicide pathway crucial during embryogenesis and tissue homeostasis. Classic example: the separation and formation of distinct digits (fingers and toes) in the human embryo occurs via apoptosis of the interdigital webbing cells!
- **Ghatnachakra High-Yield Organelle Facts**:

- **Sarcoplasmic Reticulum (SR)**: A modified smooth endoplasmic reticulum (SER) present *exclusively in muscle cells (myocytes)*; functions to sequester, store, and pump calcium ions (Ca^{2+}) for muscle contraction and relaxation.
- **Mitochondrial ATP Respiration Breakdown**: While cellular respiration theoretically generates 30–38 ATP per glucose molecule, the **terminal oxidation process / oxidative phosphorylation** (electron transport chain linked to ATP synthase in inner mitochondrial membrane) produces the vast majority—**approx. 27–34 ATP**!
- **Golgi Apparatus**: First observed in 1897 and officially reported in April 1898 by Italian physician **Camillo Golgi**; composed of stacks of membrane-bound flattened sacs called **cisternae** responsible for modifying, sorting, and packaging proteins and lipids.
- **Lysosomes ("Suicide Bags")**: Discovered by Christian de Duve; vesicular organelles packed with acid hydrolases (proteases, lipases, nucleases, carbohydrases) that function optimally in an **acidic medium (pH ~4.5–5.0)**. Functions in autophagy (digesting obsolete organelles), autolysis (cell self-destruction upon rupture), and bactericidal killing.
- **Centrosome**: The primary microtubule organizing centre (MTOC) of animal cells; contains two centrioles surrounded by pericentriolar material; organizes the spindle apparatus during mitosis. **Must-Score Fact:** Plants and most fungi lack centrosomes!
- **Lysosome vs Vacuole Taxonomic Convention**: In scientific convention, the term *lysosome* is strictly applied to animal vesicular digestive organelles, whereas in plants, fungi, and algae, these functions are performed by the large central **vacuole**.
- **Programmed Cell Death (PCD) / Apoptosis ("Cellular Suicide")**: Regulated intracellular suicide pathway crucial during embryogenesis and tissue homeostasis. Classic example: the separation and formation of distinct digits (fingers and toes) in the human embryo occurs via apoptosis of the interdigital webbing cells!
- **Plasma Membrane (Cell Membrane)**:
- **Fluid Mosaic Model (Singer & Nicolson, 1972)**: Phospholipid bilayer arranged with hydrophilic polar heads facing outward and hydrophobic non-polar fatty acid tails facing inward. Intrinsic (integral) proteins span the bilayer; extrinsic (peripheral) proteins rest on the surfaces.
- Dynamic, fluid, and **selectively permeable**: Allows water, small uncharged molecules, and selected ions to pass while blocking macromolecules.
- **Mitochondria (Chondriosome / Powerhouse of Cell)**:
- Discovered by **Albert von Kölliker (1857)** in insect muscles; named *Bioblasts* by Richard Altmann (1890); term *Mitochondria* coined by **Carl Benda (1898)**.
- **Structure**: Double membrane. Outer membrane is smooth; inner membrane is folded inward into finger-like projections called **Cristae** to dramatically increase surface area for oxidative phosphorylation.
- **Function**: Krebs cycle occurs in the fluid matrix; Electron Transport Chain (ETC) and ATP synthesis occur on the inner membrane cristae via $F_0 - F_1$ **complexes (Oxysomes / ATP synthase)**. Generates 36–38 ATP per glucose molecule.
- **Semi-autonomous nature**: Contains its own **circular double-stranded DNA (mtDNA)**, RNA, and **70S ribosomes**; replicates by binary fission.
- **Maternal Inheritance**: Mitochondrial DNA is inherited almost exclusively from the mother (via egg cytoplasm; sperm mitochondria are discarded upon fertilisation).
- **Plastids (Unique to Plant Cells & Euglenoids)**:
- Discovered by **Ernst Haeckel (1866)**; term coined by **A. F. W. Schimper (1883)**.
- Double-membrane semi-autonomous organelles with their own circular DNA and 70S ribosomes.
- **Three Major Types**:
  1. **Chloroplasts (हरितलवक)**: Green plastids containing chlorophyll (a, b) and carotenoids; kitchen of the plant cell. Internal structure has stacks of disc-like thylakoids called **Grana** (site of light reaction) surrounded by a fluid **Stroma** (site of dark reaction / Calvin cycle via RuBisCO).
  2. **Chromoplasts (वर्णीलवक)**: Yellow, orange, or red plastids containing fat-soluble carotenoid pigments (carotenes and xanthophylls). Provide colour to petals, fruits, and autumn leaves to attract pollinators: - **Lycopene**: Gives bright red colour to ripe tomatoes and watermelons. - **Capsanthin**: Red colour of ripe chillies. - **Carotene**: Orange-yellow colour of carrots. - **Betanin (Betalain)**: Purplish-red colour of beetroot. - **Anthoxanthins**: Pale yellow in flowers.
  3. **Leucoplasts (अवर्णीलवक)**: Non-pigmented, colourless storage plastids found in underground roots, tubers, and seeds: - **Amyloplasts**: Store starch / carbohydrates (e.g. potato tubers, rice grains). - **Elaioplasts / Lipoplasts**: Store oils and fats (e.g. castor seeds, groundnut). - **Aleuroplasts / Proteinoplasts**: Store proteins (e.g. maize aleurone layer).
- **Endoplasmic Reticulum (ER)**:
- Network of interconnected membranous tubules and cisternae discovered by **Keith Porter (1945)**.
- **Rough ER (RER)**: Surface studded with 80S ribosomes; extensive in cells actively engaged in protein synthesis and export (pancreatic acinar cells, plasma cells).
- **Smooth ER (SER)**: Devoid of ribosomes; major site for synthesis of lipids, phospholipids, cholesterol, and steroid hormones (testosterone, estrogen); performs **drug detoxification** in liver hepatocytes via Cytochrome P450; stores Ca^{2+} in muscle cells (**sarcoplasmic reticulum**).
- **Golgi Apparatus (Golgi Complex / Dictyosome in plants)**:
- Discovered by Italian cytologist **Camillo Golgi (1898)** using silver salt impregnation of barn owl nerve cells.
- Consists of parallel flattened sac-like **cisternae** with distinct polar faces: convex cis-face (forming / receiving face) and concave trans-face (maturing / shipping face).
- **Functions**: Packaging, modification, sorting, and secretion of proteins; post-translational **glycosylation** of proteins and lipids to form glycoproteins and glycolipids; synthesises plant cell plate pectin; forms the **acrosome** of mammalian spermatozoa and primary lysosomes.
- **Lysosomes (Suicidal Bags / Digestive Bags)**:
- Discovered by Belgian biochemist **Christian de Duve (1955)**.
- Single-membrane vesicles budded off from the trans-Golgi; contain ~50 different **acid hydrolases** (proteases, lipases, nucleases, carbohydrases) that function optimally at an **acidic pH of 4.5–5.0** maintained by proton pumps.
- **Functions**: Heterophagy (digestion of engulfed foreign pathogens, phagocytosis); Autophagy (digestion of worn-out exhausted cellular organelles); Autolysis (programmed self-destruction of damaged cells, e.g. tadpole tail resorption during metamorphosis).

- **Ribosomes (Palade Particles / Protein Factories)**:
- Discovered by Romanian-American cell biologist **George E. Palade (1953)** under the electron microscope.
- Non-membrane-bound granular particles composed of **ribosomal RNA (rRNA)** and proteins; universal to all living cells (prokaryotes and eukaryotes).
- **70S Ribosome** = 50S large subunit + 30S small subunit (found in prokaryotes, mitochondria, chloroplasts; 'S' = Svedberg unit of sedimentation coefficient).
- **80S Ribosome** = 60S large subunit + 40S small subunit (found in eukaryotic cytoplasm).
- Several ribosomes attached to a single mRNA strand form a **Polysome / Polyribosome** for simultaneous translation.
- **Centrosome & Centrioles**:
- Discovered by **Edouard van Beneden (1883)**; described by **Theodor Boveri (1888)**.
- Non-membranous organelle found in **animal cells** (absent in higher angiosperm plants).
- Contains two mutually perpendicular cylindrical structures called **Centrioles**, surrounded by pericentriolar material.
- Ultrastructure: $9 + 0$ cartwheel symmetry consisting of 9 peripheral triplets of tubulin microtubules with no central microtubule.
- **Function**: Organises the mitotic spindle apparatus during animal cell division; forms the **basal body** (kinetosome) that anchors cilia and flagella.
- **Microbodies (Single-Membrane Enzyme Sacs)**:
- **Peroxisomes**: Contain **Catalase** and peroxidase; decomposes cytotoxic hydrogen peroxide ($2H_2O_2 \rightarrow 2H_2O + O_2$); performs **photorespiration (C2 cycle)** in photosynthetic plant cells in coordination with chloroplasts and mitochondria; breaks down long-chain fatty acids via β-oxidation.
- **Glyoxysomes**: Specialized plant microbodies found in germinating fatty seeds (castor, flax, groundnut); carry enzymes of the **glyoxylate cycle** to convert stored lipids into soluble carbohydrates (gluconeogenesis) before photosynthesis begins.
- **Vacuoles & Plant Turgor**:
- Non-cytoplasmic sac bounded by a single selectively permeable membrane called the **Tonoplast**.
- In mature plant cells, the central vacuole can occupy **up to 90%** of total cell volume.
- Contains cell sap (water, mineral salts, sugars, metabolic waste, anthocyanin water-soluble pigments); maintains cell turgidity, osmotic balance, and hydrostatic pressure.

1.7 Chromosomes, DNA & RNA: The Molecular Basis of Life

- **Ghatnachakra Landmark Milestones & Transposons**:
- **Friedrich Miescher (1869)**: First discovered and isolated nucleic acid (termed *nuclein*) from the nuclei of human white blood cells (pus cells).
- **Transposable Elements ("Jumping Genes" / Transposons)**: Discovered by **Barbara McClintock** while studying pigment spot inheritance in maize (*Zea mays* kernels); awarded the Nobel Prize in Physiology or Medicine in 1983. Transposons comprise nearly **50% of the human genome**; they can mobilize, causing insertions/mutations linked to diseases like haemophilia and cancer or driving evolutionary novelty.
- **Complementary Base Pairing Suitability**: DNA's double-stranded antiparallel structure with specific base pairing ($A = T$ with 2 hydrogen bonds, $G \equiv C$ with 3 hydrogen bonds) makes it uniquely suited to store and faithfully transmit genetic information across generations, and allows repair enzymes to recognize and replace damaged bases using the opposite strand as an intact template.
- **The Genetic Code (Universal & Commaless)**: Deciphered by Nirenberg, Khorana, and Holley (Nobel 1968). Consists of **64 triplet codons** coding for 20 standard amino acids:
  - **61 Sense Codons**: Code for specific amino acids.
  - **1 Start Codon**: **AUG** (also codes for Methionine).
  - **3 Stop / Nonsense Codons**: **UAA, UAG, UGA** (do not code for any amino acid; trigger translation termination).
  - **Properties**: Universal across virtually all life, commaless (continuous reading frame), non-overlapping, non-ambiguous (one codon specifies only one amino acid), and degenerate/redundant (multiple codons specify the same amino acid).
- **Ribozymes (Catalytic RNA)**: RNA molecules capable of acting as biological catalysts (enzymes), discovered by Thomas Cech and Sidney Altman (Nobel Prize in Chemistry 1989).

- **Ghatnachakra Landmark Milestones & Transposons**:
- **Friedrich Miescher (1869)**: First discovered and isolated nucleic acid (termed *nuclein*) from the nuclei of human white blood cells (pus cells).
- **Transposable Elements ("Jumping Genes" / Transposons)**: Discovered by **Barbara McClintock** while studying pigment spot inheritance in maize (*Zea mays* kernels); awarded the Nobel Prize in Physiology or Medicine in 1983. Transposons comprise nearly **50% of the human genome**; they can mobilize, causing insertions/mutations linked to diseases like haemophilia and cancer or driving evolutionary novelty.
- **Complementary Base Pairing Suitability**: DNA's double-stranded antiparallel structure with specific base pairing ($A = T$ with 2 hydrogen bonds, $G \equiv C$ with 3 hydrogen bonds) makes it uniquely suited to store and faithfully transmit genetic information across generations, and allows repair enzymes to recognize and replace damaged bases using the opposite strand as an intact template.
- **The Genetic Code (Universal & Commaless)**: Deciphered by Nirenberg, Khorana, and Holley (Nobel 1968). Consists of **64 triplet codons** coding for 20 standard amino acids:
  - **61 Sense Codons**: Code for specific amino acids.
  - **1 Start Codon**: **AUG** (also codes for Methionine).
  - **3 Stop / Nonsense Codons**: **UAA, UAG, UGA** (do not code for any amino acid; trigger translation termination).
  - **Properties**: Universal across virtually all life, commaless (continuous reading frame), non-overlapping, non-ambiguous (one codon specifies only one amino acid), and degenerate/redundant (multiple codons specify the same amino acid).
- **Ribozymes (Catalytic RNA)**: RNA molecules capable of acting as biological catalysts (enzymes), discovered by Thomas Cech and Sidney Altman (Nobel Prize in Chemistry 1989).

- **Chromosome Structure & Discovery**:
- First observed by Hofmeister (1848); named "Chromosome" (coloured body) by **Wilhelm von Waldeyer (1888)**.
- Chromatin consists of **DNA (~31%)**, **Histone proteins (~36%)**, non-histone proteins (~28%), and RNA (~5%). Histones are rich in basic amino acids **Lysine and Arginine** and carry a net positive charge that binds tightly to negatively charged DNA phosphate groups.
- **Nucleosome Model (Kornberg & Thomas, 1974)**: Structural repeating unit of chromatin; 146 base pairs of DNA wrapped around an octamer of 8 histone molecules ($2 \times \mathbf{H2A, H2B, H3, H4}$) with H1 acting as linker histone ("beads-on-a-string").
- **Chromosomal Anatomy**:
  - **Centromere (Primary Constriction)**: Site of spindle fiber attachment via disc-shaped protein complexes called **Kinetochores**.
  - **Classification by Centromere Position**: 1. **Metacentric**: Centromere exactly in the middle; equal arms; V-shaped during anaphase. 2. **Sub-metacentric**: Centromere slightly off-centre; one shorter arm (p-arm) and one longer arm (q-arm); L-shaped during anaphase. 3. **Acrocentric**: Centromere near one end; one extremely short arm; J-shaped during anaphase. 4. **Telocentric**: Centromere at the terminal tip; single arm; I-shaped during anaphase.
  - **Telomeres**: Heterochromatic protective caps at chromosome ends containing repeated hexanucleotide sequences ($TTAGGG$ in humans) synthesised by the enzyme **Telomerase**; prevent chromosome fusion and degradation; shorten with each somatic cell division (cellular biological clock of ageing).
  - **SAT Chromosome (Satellite)**: Chromosome with a secondary constriction beyond which lies a non-staining knob-like fragment called a **satellite** (acts as Nucleolar Organizer Region / NOR).
- **Giant Chromosomes (Cytological Curiosities)**:
- **Polytene Chromosomes (Salivary Gland Chromosomes)**: Discovered by **E. G. Balbiani (1881)** in salivary gland cells of *Chironomus* midge larvae; formed by repeated rounds of DNA replication without cell division (**Endomitosis**); show prominent uncoiled puffs or **Balbiani rings** which are sites of intense RNA synthesis / transcription.
- **Lampbrush Chromosomes**: Discovered by **Walther Flemming (1882)** and J. Rückert (1892) in growing oocytes of amphibians, sharks, and birds during the prolonged diplotene stage of meiosis; show lateral projecting loops of uncoiled DNA resembling a test-tube cleaning brush; active in yolk synthesis and mRNA transcription.
- **DNA (Deoxyribonucleic Acid) Structure**:
- **Watson & Crick Double Helix Model (1953)**:
  - Based on X-ray diffraction data of B-DNA captured by **Rosalind Franklin** (Photograph 51) and Maurice Wilkins.
  - Two anti-parallel polynucleotide chains: one runs $5' \rightarrow 3'$, the other runs $3' \rightarrow 5'$.
  - Backbone consists of alternating **deoxyribose sugars** and **phosphate groups** linked by covalent $3'-5'$ **phosphodiester bonds**.
  - Hydrophobic nitrogenous bases face inward, perpendicular to the helix axis, linked by weak **hydrogen bonds**.
  - **Purines** (double-ring: **Adenine [A]** and **Guanine [G]**) pair strictly with **Pyrimidines** (single-ring: **Thymine [T]** and **Cytosine [C]**):
  - $A = T$ (linked by 2 hydrogen bonds).
  - $G \equiv C$ (linked by 3 hydrogen bonds; higher GC content = higher melting temperature T_m).
  - **Dimensions of B-DNA**:
  - Diameter of helix = $20 \text{ \AA}$ **(2.0 nm)**.
  - Pitch / Length of one complete helical turn = $34 \text{ \AA}$ **(3.4 nm)**.
  - Distance between two adjacent base pairs = $3.4 \text{ \AA}$ **(0.34 nm)**.
  - Number of base pairs per complete helical turn = 10 **bp**.
- **Chargaff's Equimolar Rules (Erwin Chargaff, 1950)**:
- In any double-stranded DNA molecule:
  1. Amount of Adenine equals Thymine ($[A] = [T]$) and Guanine equals Cytosine ($[G] = [C]$).
  2. Total purines equal total pyrimidines: $[A + G] = [T + C]$, hence $\frac{A+G}{T+C} = 1$.
  3. The base ratio $\frac{A+T}{G+C}$ is constant for a given species, but varies widely between different species.
- **RNA (Ribonucleic Acid) Structure & Classes**:
- Single-stranded polymer of ribonucleotides; pentose sugar is **Ribose** (carries an extra reactive $-OH$ group at the $2'$ carbon, making RNA less stable and more reactive than DNA).
- Nitrogenous bases: Purines (**Adenine**, **Guanine**) and Pyrimidines (**Cytosine**, **Uracil**; Uracil replaces Thymine: Thymine is chemically 5-methyl uracil).
- **Three Main Types of Non-Genetic RNA**:
  1. **mRNA (Messenger RNA, 5–10% of total RNA)**: Carries the genetic message transcribed from nuclear DNA to ribosomes in the cytoplasm; read in triplets called **Codons**.
  2. **tRNA (Transfer RNA / Soluble RNA / Adaptor RNA, 10–15%)**: Smallest RNA ($70 - 90$ nucleotides); cloverleaf 2D model (Robert Holley, 1965) and L-shaped 3D structure (Kim, 1973); carries specific amino acids at its $3'$-CCA terminal end and binds to mRNA via its complementary **Anticodon loop**.
  3. **rRNA (Ribosomal RNA, 80% of total RNA)**: Most abundant, stable, and long-lived RNA; structural component of ribosomes; catalytic functions (e.g. 23S rRNA in bacteria acts as **Peptidyl transferase / Ribozyme** to form peptide bonds without protein enzymes).
- **Central Dogma of Molecular Biology**:
- Proposed by **Francis Crick (1958)**: Unidirectional flow of genetic information:

$$\text{DNA} \xrightarrow[\text{Transcription}]{\text{RNA Polymerase}} \text{mRNA} \xrightarrow[\text{Translation}]{\text{Ribosomes}} \text{Protein}$$

- **Reverse Transcription (Central Dogma Reverse / Teminism)**:
  - Discovered by **Howard Temin and David Baltimore (1970)** in retroviruses (Rous Sarcoma Virus):

$$\text{RNA} \xrightarrow[\text{Reverse Transcriptase}]{\text{RNA-dependent DNA Polymerase}} \text{Complementary DNA (cDNA)}$$

- **The Genetic Code & Codons**:
- Deciphered by **Marshall Nirenberg, Heinrich Matthaei, and Har Gobind Khorana** (Nobel Prize in 1968).
- Triplet code: 64 codons total ($4^3 = 64$ combinations) coding for 20 standard amino acids.
- Universal, unambiguous (one codon codes for only one amino acid), degenerate (most amino acids are coded by more than one codon), and comma-less.
- **Start Codon**: **AUG** (codes for **Methionine** in eukaryotes; formyl-methionine in prokaryotes).
- **Stop Codons (Nonsense Codons / Termination Codons)**: **UAA** (Ochre), **UAG** (Amber), and **UGA** (Opal) — do not code for any amino acid; cause termination of protein synthesis.

1.8 Cell Cycle & Cell Division (Mitosis & Meiosis)

- **The Eukaryotic Cell Cycle**:
- Alternation of **Interphase** (resting/preparatory phase, occupying ~95% of total cycle duration) and **M-Phase** (Mitosis / division phase, ~5% duration).
- **Phases of Interphase**:
  1. **G_1 Phase (Gap 1 / Post-mitotic gap)**: Rapid synthesis of RNA, enzymes, and structural proteins; cell volume increases; chromosome remains single-stranded ($2n, 2C$).
  2. **G_0 Phase (Quiescent Stage)**: Cells withdraw from the cell cycle and cease to divide while remaining metabolically active (e.g. adult human neurons, cardiac myocytes).
  3. **S Phase (Synthetic Phase)**: **DNA replication** occurs strictly in this phase; nuclear DNA content doubles from $2C \rightarrow 4C$, while chromosome number remains unchanged at $2n$; centrioles duplicate in the cytoplasm of animal cells.
  4. **G_2 Phase (Gap 2 / Pre-mitotic gap)**: Synthesis of spindle tubulin proteins and ATP; cell prepares to enter mitosis.
- **Mitosis (Equational Division / समसूत्री विभाजन)**:
- First observed by Strasburger (1870) in plants; term coined by German anatomist **Walther Flemming (1882)** from animal cells.
- Occurs in **somatic cells** for vegetative growth, tissue repair, and asexual reproduction. Chromosome number in daughter cells remains identical to the parent cell ($2n \rightarrow 2n$).
- **Karyokinesis (Nuclear Division Stages)**:
  1. **Prophase**: Chromatin condenses into distinct chromosomes; nucleolus and nuclear membrane disintegrate; centrosomes move to opposite poles.
  2. **Metaphase**: Nuclear envelope completely disappears; chromosomes are shortest and thickest; align at the equatorial plane (**Metaphase plate**); spindle fibers attach to the **kinetochores** of centromeres. - **Logic: Metaphase is the best stage to study chromosome morphology, size, and count.**
  3. **Anaphase**: Centromeres split simultaneously; sister chromatids separate into individual daughter chromosomes and migrate toward opposite poles pulled by spindle shortening. - **Logic: Anaphase is the best stage to study chromosome shapes (V, L, J, I).**
  4. **Telophase**: Chromosomes decondense back into chromatin at poles; nuclear membrane and nucleoli reform; spindle apparatus disassembles.
- **Cytokinesis (Cytoplasmic Division)**:
  - Animal cells: **Cleavage furrow** method (plasma membrane constricts from outside inward, centripetal).
  - Plant cells: **Cell plate** method (phragmoplast formed by Golgi vesicles grows from centre outward, centrifugal, forming middle lamella of calcium/magnesium pectate).
- **Mitotic Inhibitors & Polyploidy**:
  - **Colchicine**: Alkaloid extracted from the corms of meadow saffron (*Colchicum autumnale*); binds to tubulin dimers and completely inhibits spindle fiber formation, preventing anaphase separation and arresting dividing cells at **metaphase**; induces artificial **polyploidy** ($2n \rightarrow 4n, 8n$).

```
Meiosis-I: Prophase-I Detailed Sub-Stages (Mnemonic: Lazy Zebras Paint Dirty Dirt)
Leptotene ──► Zygotene ──────────► Pachytene ──────────────► Diplotene ──────► Diakinesis
(Bouquet)     (Synapsis;            (CROSSING OVER;           (Chiasmata;        (Terminalisation;
               Synaptonemal          Recombinase Enzyme;       Synaptonemal       Spindle Forms)
               Complex)              Tetrad Visible)           Dissolves)
```

- **Meiosis (Reductional Division / अर्धसूत्री विभाजन)**:
- Term coined by **J. B. Farmer and J. E. S. Moore (1905)**.
- Occurs strictly in diploid reproductive cells (**gametocytes / meiocytes**) during gametogenesis (spermatogenesis and oogenesis), yielding **4 genetically distinct haploid (n) daughter cells**.
- Comprises two sequential nuclear divisions with only a single cycle of DNA replication:
  - **Meiosis-I**: Heterotypic / **Reductional division** (homologous chromosome pairs separate; chromosome number is halved from $2n \rightarrow n$).
  - **Meiosis-II**: Homotypic / **Equational division** (identical to mitosis; sister chromatids separate; chromosome number remains $n \rightarrow n$).
- **Detailed Sub-stages of Prophase-I (Longest and Most Complex Phase)**:
  1. **Leptotene (लेप्टोटीन)**: Chromatin condenses into long, slender visible threads; chromosomes show a "bouquet stage" as telomeres converge toward the centrosome.
  2. **Zygotene (जाइगोटीन)**: Homologous chromosomes find each other and pair up gene-for-gene — this pairing process is called **Synapsis**; a tripartite protein ladder structure called the **Synaptonemal Complex** forms between them; paired unit is called a **Bivalent** or **Tetrad**.

3. **Pachytene (पैकाइटीन — CRITICAL STAGE)**: - Non-sister chromatids of homologous bivalents exchange corresponding genetic segments — this phenomenon is called **Crossing Over**. - Catalysed by the multi-enzyme complex **Recombinase** (endonuclease + ligase). - Generates **new genetic recombinations**, providing the raw material for organic evolution.
4. **Diplotene (डिप्लोटीन)**: - Dissolution of the synaptonemal complex; homologous chromosomes begin to repel each other and pull apart, remaining attached only at points where crossing over occurred. - These X-shaped physical contact points are called **Chiasmata** (singular: chiasma). - In oocytes of certain vertebrates (amphibians, humans), diplotene can remain arrested for months or years (**Dictyotene stage**).
5. **Diakinesis (डायाकाइनेसिस)**: - Complete **Terminalisation of chiasmata** (chiasmata shift toward chromosome tips like an unzipping zipper); nucleolus and nuclear envelope disappear; spindle apparatus forms.

1.9 Heredity & Mendelian Genetics

- **Ghatnachakra Genetics & Chromosome Facts**:
- **Gregor Johann Mendel**: Augustinian cleric who conducted hybridization experiments on garden pea (*Pisum sativum*) at the St. Thomas's Abbey monastery in Brünn (now Brno, Czech Republic). Mendel established that inheritance occurs exclusively via **sexual reproduction** and equal biparental genetic contribution.
- **Chromosomal Theory of Inheritance**: Proposed independently by **Walter Sutton and Theodor Boveri (1902–1903)**, noting the exact parallelism between Mendelian factor segregation and homologous chromosome separation during meiosis. Later confirmed experimentally in *Drosophila* by **Thomas Hunt Morgan (1915)**.
- **Human Chromosomal Karyotype**: 23 pairs (46 total chromosomes) = 22 pairs of autosomes (44) + 1 pair of allosomes/sex chromosomes (XX female, XY male). The male parent's sperm (contributing either X or Y) solely determines the sex of the offspring.
- **Reptilian Sex Chromosome Exception**: Many species of lizards (such as the genus *Sceloporus*) lack morphologically recognizable sex chromosomes; sex in many reptiles is determined environmentally by incubation temperature (Temperature-dependent Sex Determination / TSD).
- **Occupational Radiation Risk**: **X-ray technicians** and medical imaging professionals face the highest occupational hazard of cumulative ionizing radiation causing double-strand breaks and permanent mutagenic alterations in cellular DNA.
- **Ghatnachakra Genetics & Chromosome Facts**:
- **Gregor Johann Mendel**: Augustinian cleric who conducted hybridization experiments on garden pea (*Pisum sativum*) at the St. Thomas's Abbey monastery in Brünn (now Brno, Czech Republic). Mendel established that inheritance occurs exclusively via **sexual reproduction** and equal biparental genetic contribution.
- **Chromosomal Theory of Inheritance**: Proposed independently by **Walter Sutton and Theodor Boveri (1902–1903)**, noting the exact parallelism between Mendelian factor segregation and homologous chromosome separation during meiosis. Later confirmed experimentally in *Drosophila* by **Thomas Hunt Morgan (1915)**.
- **Human Chromosomal Karyotype**: 23 pairs (46 total chromosomes) = 22 pairs of autosomes (44) + 1 pair of allosomes/sex chromosomes (XX female, XY male). The male parent's sperm (contributing either X or Y) solely determines the sex of the offspring.
- **Reptilian Sex Chromosome Exception**: Many species of lizards (such as the genus *Sceloporus*) lack morphologically recognizable sex chromosomes; sex in many reptiles is determined environmentally by incubation temperature (Temperature-dependent Sex Determination / TSD).
- **Occupational Radiation Risk**: **X-ray technicians** and medical imaging professionals face the highest occupational hazard of cumulative ionizing radiation causing double-strand breaks and permanent mutagenic alterations in cellular DNA.
- **Gregor Johann Mendel (1822–1884)**:
- Austrian monk, Father of Genetics; conducted hybridisation experiments on the garden pea (***Pisum sativum***) for 7 years (**1856–1863**) at the St. Thomas Monastery in Brünn (Brno).
- Published findings in 1866: *"Experiments in Plant Hybridisation"* in the proceedings of the Natural History Society of Brünn.
- Work remained unrecognised until **1900**, when it was independently rediscovered by **Hugo de Vries** (Holland), **Carl Correns** (Germany), and **Erich von Tschermak** (Austria).
- **Reasons for Mendel's Choice of Garden Pea**: Well-defined contrasting traits, annual life cycle, bisexual flowers, naturally self-pollinating but easily cross-pollinated manually, produces abundant fertile offspring.
- **Mendel's 7 Pairs of Contrasting Characters in Garden Pea**:

Character	Dominant Trait (प्रभावी)	Recessive Trait (अप्रभावी)	Chromosome Location
Stem Height	Tall (T)	Dwarf (t)	Chromosome 4
Flower Colour	Violet / Purple (W)	White (w)	Chromosome 1
Flower Position	Axial (A)	Terminal (a)	Chromosome 4
Pod Shape	Inflated / Full (I)	Constricted (i)	Chromosome 4
Pod Colour	Green (G)	Yellow (g)	Chromosome 5
Seed Shape	Round (R)	Wrinkled (r)	Chromosome 7
Seed Colour (Cotyledon)	Yellow (Y)	Green (y)	Chromosome 1

- **Mendelian Principles & Laws**: 1. **Principle of Paired Factors**: Characters are controlled by discrete units called factors (now called **Genes**, term coined by **Wilhelm Johannsen, 1909**); genes occur in pairs called **Alleles** (alternative forms of the same gene occupying identical loci on homologous chromosomes). 2. **Law of Dominance (प्रभाविता का नियम)**: In a monohybrid cross between two pure contrasting parents, only one ancestral trait appears in the F_1 generation (the **Dominant** trait); the hidden trait that fails to express is the **Recessive** trait. 3. **Law of Segregation (विपृथक्करण का नियम / Law**

of Purity of Gametes): The two alleles of a gene do not blend in a hybrid, but separate and segregate from each other during gamete formation so that each gamete receives only one allele with equal probability.

- **Logic: This law is universal without any exceptions.** 4. **Law of Independent Assortment (स्वतंत्र अपव्यूहन का नियम)**: In a dihybrid cross, the segregation of alleles of one gene pair is completely independent of the segregation of alleles of another gene pair during gamete formation.
- **Logic: Applies only to genes situated on different chromosomes or situated far apart on the same chromosome; violated by Linkage.**

- **Key Genetic Ratios for Competitive Prelims**:
- **Monohybrid Cross ($Tt \times Tt$)**:
  - Phenotypic ratio = $3:1$ (3 Tall : 1 Dwarf).
  - Genotypic ratio = $1:2:1$ ($1\,TT:2\,Tt:1\,tt$).
- **Dihybrid Cross ($RrYy \times RrYy$)**:
  - Phenotypic ratio = $9:3:3:1$ (9 Round Yellow : 3 Round Green : 3 Wrinkled Yellow : 1 Wrinkled Green).
  - Genotypic ratio = $1:2:1:2:4:2:1:2:1$.
- **Test Cross**: Cross of an organism showing dominant phenotype with its **homozygous recessive parent** ($T? \times tt$) to determine whether the dominant individual is homozygous (TT) or heterozygous (Tt):
  - Monohybrid test cross ratio = $1:1$.
  - Dihybrid test cross ratio = $1:1:1:1$.

- **Non-Mendelian Genetics & Extensions**:
- **Incomplete Dominance (अपूर्ण प्रभाविता)**: Discovered by **Carl Correns (1903)** in 4 o'clock plant (*Mirabilis jalapa*) and snapdragon (*Antirrhinum majus*). Neither allele is fully dominant; the heterozygous F_1 shows an intermediate blended phenotype:
  - Pure Red (RR) × Pure White (rr) → All **Pink (Rr)**.
  - F_2 Phenotypic ratio = $1:2:1$ (1 Red : 2 Pink : 1 White) — **Phenotypic and genotypic ratios are identical**.
- **Codominance (सह-प्रभाविता)**: Both alleles of a gene express themselves simultaneously and fully in the heterozygote without blending:
  - Classic example: **Human ABO Blood Groups**. Allele I^A produces A-antigen; allele I^B produces B-antigen. When both are present (I^AI^B), the individual has **Type AB blood** expressing both antigens.
  - Coat colour in cattle (Roan colour = mix of red and white hairs).
- **Multiple Allelism**: When a single gene exists in more than two alternative allelic forms within a population. Human ABO blood groups are controlled by **3 alleles (I^A, I^B, i)**, which produce **6 genotypes** and **4 phenotypes** (A, B, AB, O). Number of genotypes = $\frac{n(n+1)}{2} = \frac{3(4)}{2} = 6$.
- **Pleiotropy (एकप्रभाविता)**: A single gene simultaneously influences multiple, seemingly unrelated phenotypic traits:
  - **Phenylketonuria (PKU)**: Mutation in the phenylalanine hydroxylase gene causes mental retardation, reduced hair and skin pigmentation.
  - **Sickle-cell anaemia**: Single β-globin gene mutation causes abnormal sickle RBCs, splenic infarction, joint pain, and kidney failure.
- **Polygenic Inheritance (Quantitative Inheritance)**: A single phenotypic trait is controlled by the cumulative additive effect of three or more independent genes:
  - Examples: **Human skin colour** (described by Davenport using alleles A, B, C), **Human height**, **Intelligence**, **Kernel colour in wheat** (Nilsson-Ehle). Produces a continuous bell-shaped Gaussian curve.
- **Linkage (सहलग्नता)**:
  - Discovered and established by **Thomas Hunt Morgan (1910)** working on the fruit fly (*Drosophila melanogaster*).
  - Tendency of two or more genes located on the same chromosome to stay together and be inherited together into the same gamete without independent assortment.
  - Violates Mendel's Law of Independent Assortment. Strength of linkage is inversely proportional to the physical distance between genes on the chromosome.

- **Sex Determination Mechanisms**:
- **$XX - XY$ System**: Humans, most mammals, and *Drosophila*. Females are homogametic (XX); males are **heterogametic (XY)**. The male father determines the sex of the child (sperm carries either X or Y with 50% probability).
- **$ZZ - ZW$ System**: Birds, reptiles, and butterflies. Males are homogametic (ZZ); females are **heterogametic (ZW)**. The female mother determines the sex of the offspring.
- **$XX - XO$ System**: Grasshoppers and roundworms. Females have two X chromosomes (XX); males have only one X chromosome (XO).
- **Haplodiploidy (Honeybees, Ants, Wasps)**:
  - Sex is determined by the number of chromosome sets:
  - Fertilised egg ($2n = 32$) develops into **diploid females** (Queen or sterile Worker bees).
  - Unfertilised egg ($n = 16$) develops into **haploid males (Drones)** by **Parthenogenesis**.
  - Drones have no father and cannot have sons, but they have grandfathers and can have grandsons.

- **Barr Body & Lyon's Hypothesis (Dosage Compensation)**:
- Formulated by British geneticist **Mary Lyon (1961)**.
- In female mammalian somatic cells, one of the two X chromosomes is randomly and permanently inactivated during early embryonic development, condensing into a dark heterochromatic mass called the **Barr Body** (visible against the nuclear membrane).
- **Formula**: $\text{Number of Barr bodies} = \text{Total X chromosomes} - 1$.
- Normal male ($46, XY$) = 0 Barr bodies.
- Normal female ($46, XX$) = 1 Barr body.

- Turner syndrome female ($45, XO$) = 0 Barr bodies.
- Klinefelter syndrome male ($47, XXY$) = 1 Barr body.
- Super-female ($47, XXX$) = 2 Barr bodies.

1.10 Mutation & Human Genetic Disorders

- **Mutation & Its Mechanisms**:
- Term coined by Dutch botanist **Hugo de Vries (1901)** in his book *The Mutation Theory*, working on evening primrose (*Oenothera lamarckiana*).
- Mutations are sudden, stable, discontinuous, and heritable changes in the nucleotide sequence of DNA.
- De Vries stated that evolution occurs through large, single-step mutations called **Saltation**, in contrast to Darwin's gradual minor variations.
- **Classification of Mutations**:
  1. **Point Mutation / Gene Mutation**: Change in a single base pair of DNA: - **Transition**: Purine replaced by purine ($A \leftrightarrow G$) or pyrimidine by pyrimidine ($C \leftrightarrow T$). - **Transversion**: Purine replaced by pyrimidine ($A/G \leftrightarrow C/T$) or vice versa. - **Missense / Non-synonymous mutation**: Alters codon to code for a different amino acid (e.g. Sickle-cell anaemia). - **Nonsense mutation**: Changes an amino acid codon into a premature stop codon (UAA, UAG, UGA), truncating the protein. - **Silent / Synonymous mutation**: Base changes but codes for the same amino acid due to degeneracy.
  2. **Frameshift Mutation**: Insertion or deletion of 1 or 2 nucleotides (not multiples of three) that shifts the entire translational reading frame from that point onward.
  3. **Chromosomal Aberrations**: Structural alterations in chromosomes: - **Deletion**: Loss of a chromosomal segment (e.g. Cri-du-chat syndrome). - **Duplication**: Repetition of a segment. - **Inversion**: A segment breaks, rotates $180°$, and rejoins. - **Translocation**: Exchange of segments between non-homologous chromosomes (e.g. Philadelphia chromosome: reciprocal translocation between chromosomes 9 and 22, linked with Chronic Myeloid Leukemia / CML).
- **Mutagens (Mutation-Inducing Agents)**:
  - Physical mutagens: Ionising radiation (X-rays, gamma rays; H. J. Muller 1927 first induced mutations in *Drosophila* with X-rays); Non-ionising radiation (UV rays, forms thymine dimers).
  - Chemical mutagens: Mustard gas, nitrous acid (HNO_2), 5-bromouracil (base analog), ethidium bromide, acridine orange.
- **Human Genetic Disorders**:
- Divided into **Mendelian Disorders** (single-gene defects) and **Chromosomal Disorders** (aneuploidy or polyploidy).
- **Major Mendelian Disorders (Tested heavily in PSCs)**:

Disorder	Inheritance Pattern	Molecular Cause / Biochemical Defect	Clinical Symptoms
Sickle-cell Anaemia	Autosomal Recessive (Chromosome 11)	Point mutation replacing **Glutamic acid with Valine** at position 6 of β-globin chain ($GAG \rightarrow GUG$)	RBCs become sickle-shaped under low O_2, block capillaries; heterozygotes ($Hb^A Hb^S$) are resistant to Falciparum malaria
Haemophilia (Bleeder's Disease)	X-linked Recessive	Deficiency of blood clotting **Factor VIII** (Haemophilia A, 80%) or **Factor IX** (Haemophilia B / Christmas disease)	Prolonged, non-stop bleeding from minor cuts; famous in European royal families tracing to Queen Victoria
Colour Blindness (Daltonism)	X-linked Recessive	Defect in red or green cone photopigments of the retina	Inability to distinguish red and green; affects ~8% of human males and only ~0.4% of females
Phenylketonuria (PKU)	Autosomal Recessive (Chromosome 12)	Lack of the hepatic enzyme **Phenylalanine Hydroxylase**, causing phenylalanine accumulation	Phenylalanine converts to phenylpyruvic acid; accumulates in brain causing severe mental retardation; excreted in urine
Thalassemia	Autosomal Recessive (Chr 16 for α, Chr 11 for β)	**Quantitative defect**: Reduced synthesis of normal α- or β-globin peptide chains	Microcytic hypochromic anaemia, splenomegaly, bone deformities; requires lifelong blood transfusions
Albinism	Autosomal Recessive (Chromosome 11)	Deficiency of copper-containing enzyme **Tyrosinase**; blocks conversion of DOPA to melanin	Complete absence of dark melanin pigment in skin, hair, and iris of the eyes; extreme photophobia
Huntington's Disease	**Autosomal Dominant** (Chromosome 4)	Unstable CAG trinucleotide repeat expansion in huntingtin gene	Adult-onset progressive neurodegeneration, involuntary choreic movements, dementia; fatal
Cystic Fibrosis	Autosomal Recessive (Chromosome 7)	Mutation in CFTR gene (chloride channel)	Thick sticky mucus clogs lungs and pancreatic ducts; chronic respiratory infections

- **Major Chromosomal Disorders (Aneuploidies from Non-Disjunction)**:
- **Down Syndrome (Mangolism / Trisomy 21)**:
  - First described by British physician **John Langdon Down (1866)**.
  - Karyotype: $47, 21+1$ (an extra copy of chromosome 21 due to maternal meiotic non-disjunction; risk rises steeply with advanced maternal age).
  - Symptoms: Short stature, small round head, furrowed tongue with partially open mouth, broad palm with single deep transverse **simian crease**, congenital heart disease, mental retardation.
- **Turner Syndrome ($45, XO$)**:
  - Discovered by **Henry Turner (1938)**.
  - Karyotype: $45, XO$ (monosomy of X chromosome; completely lacks one sex chromosome).

- Symptoms: Phenotypically female, sterile, rudimentary streak ovaries, short stature, webbed neck, broad shield-shaped chest with widely spaced nipples, absence of secondary sexual characters, **0 Barr bodies**.

- **Klinefelter Syndrome ($47, XXY$)**:
  - Discovered by **Harry Klinefelter (1942)**.
  - Karyotype: $47, XXY$ (an extra X chromosome in phenotypic males).
  - Symptoms: Tall stature with disproportionately long lower limbs, sterile male, testicular atrophy, sparse body hair, development of breast tissue (**Gynaecomastia**), high-pitched feminine voice, **1 Barr body**.

- **Cri-du-chat (Cat's Cry) Syndrome**:
  - Discovered by Jérôme Lejeune (1963).
  - Cause: Partial **deletion of the short arm (p-arm) of chromosome 5** ($46, 5p-$).
  - Symptoms: Infant cries with a high-pitched sound resembling a mewing cat (due to malformed larynx), microcephaly, wide-set eyes, severe intellectual disability.

- **Edward Syndrome (Trisomy 18)**: Extra chromosome 18; clenched fists with overlapping fingers, rocker-bottom feet.
- **Patau Syndrome (Trisomy 13)**: Extra chromosome 13; cleft lip, cleft palate, polydactyly, severe mental defect.

1.11 Genetic Engineering, Biotechnology & Applied Modern Tools

- **Recombinant DNA (rDNA) Technology Foundations**:
- The construction of the first artificial recombinant DNA molecule was achieved by **Paul Berg (1972)** (Father of Genetic Engineering; introduced SV40 virus gene into *E. coli* using lambda phage; Nobel Prize in 1980).
- **Stanley Cohen and Herbert Boyer (1972–1973)** isolated an antibiotic-resistance gene from a plasmid of *Salmonella typhimurium* and linked it to a plasmid vector of *E. coli* using restriction enzymes and DNA ligase.
- **Core Molecular Tools of Genetic Engineering**: 1. **Restriction Endonucleases (Molecular Scissors / आणविक कैंची)**:
  - Discovered by **Stewart Linn, Werner Arber, Hamilton Smith, and Daniel Nathans** (Nobel Prize in 1978).
  - Enzymes that recognise specific palindromic nucleotide sequences ($4 - 8$ bp) in double-stranded DNA and cleave both strands at specific phosphodiester backbone points:
  - **Palindromic sequence**: Reads identical forward ($5' \rightarrow 3'$) and backward ($3' \rightarrow 5'$), e.g. $5' - GAATTC - 3'$ for the enzyme ***EcoRI*** (isolated from *Escherichia coli* strain RY13).
  - First discovered restriction endonuclease was ***HindII*** (cleaves at a specific 6 bp recognition site).
  - Can produce sticky / cohesive staggered ends (*EcoRI*, *BamHI*) or blunt ends (*SmaI*, *HaeIII*). 2. **DNA Ligase (Molecular Glue / आणविक गोंद)**:
  - Discovered by Gellert (1967); seals single-stranded nicks by catalysing covalent phosphodiester bond formation between adjacent $3'$-OH and $5'$-$phosphate$ ends of DNA. 3. **Cloning Vectors (Gene Carriers / वाहन)**:
  - DNA molecules used as vehicles to artificially carry foreign genetic material into a host cell for replication:
  - **Plasmids**: Autonomously replicating, extra-chromosomal, circular double-stranded DNA molecules found naturally in bacteria (e.g. pBR322, pUC19).
  - ***Agrobacterium tumefaciens***: Known as the **"Natural Genetic Engineer of Plants"**. Contains the **Ti Plasmid (Tumour-inducing plasmid)** that naturally transfers a segment of DNA (T-DNA) into plant host cells causing crown gall tumours; modified by disarming its tumour genes to act as the primary vector for plant transformation.
  - Other vectors: Bacteriophages (λ phage), Cosmids (plasmid + cos site), BAC (Bacterial Artificial Chromosomes), YAC (Yeast Artificial Chromosomes, carries largest DNA inserts up to 1000 kb). 4. **Polymerase Chain Reaction (PCR / आणविक फोटोकॉपी)**:
  - Invented by American biochemist **Kary B. Mullis (1983)** (Nobel Prize in Chemistry, 1993).
  - *In vitro* enzymatic amplification of a specific DNA segment into billions of copies within hours.
  - **Three Essential Steps in One PCR Cycle**: 1. **Denaturation ($94°–96°C$)**: High temperature breaks hydrogen bonds, separating double-stranded DNA into two single strands. 2. **Annealing ($50°–60°C$)**: Two synthetic oligonucleotide primers hybridise to their complementary sequences at the $3'$ ends of single-stranded DNA templates. 3. **Extension ($72°C$)**: DNA polymerase synthesises the new complementary strand by adding dNTPs.
  - **Crucial Enzyme**: ***Taq* DNA Polymerase**, a heat-stable enzyme isolated from the thermophilic bacterium ***Thermus aquaticus*** (native to hydrothermal hot springs; remains active at $95°C$). 5. **Agarose Gel Electrophoresis**:
  - Separates DNA or RNA fragments according to their **molecular size and charge**.
  - DNA is **negatively charged** (due to phosphate groups); under an electric field, fragments migrate through the porous agarose matrix toward the **positive electrode (Anode)**; smaller fragments move faster and travel further.
  - Separated bands cannot be seen directly; stained with **Ethidium Bromide (EtBr)** and visualised under **UV light**, fluorescing as **bright orange bands**. 6. **Blotting Techniques for Macromolecule Detection**:
  - **Southern Blotting**: Developed by **Edward M. Southern (1975)** for identifying specific **DNA** fragments using labeled complementary probes.
  - **Northern Blotting**: Developed by Alwine (1977) for identifying specific **RNA** sequences.
  - **Western Blotting**: Developed by George Stark and Harry Towbin (1979) for identifying specific **Proteins** using labeled antibodies.
  - *Mnemonic*: **SNOW DROP** (Southern-DNA, Northern-RNA, o-o, Western-Protein). 7. **DNA Fingerprinting (DNA Profiling)**:
  - Developed by British geneticist **Sir Alec Jeffreys (1984)**; introduced in India by **Dr. Lalji Singh** and **Dr. V. K. Kashyap** (Lalji Singh is known as the Father of Indian DNA Fingerprinting).
  - Identifies individuals based on unique variations in non-coding repetitive DNA sequences called **VNTRs (Variable Number Tandem Repeats)** or Minisatellites.
  - Used in forensic criminology, paternity dispute resolution, and evolutionary lineage tracking. 8. **Hybridoma Technology & Monoclonal Antibodies (mAbs)**:

- Developed by German immunologist **Georges Köhler and César Milstein (1975)** at Cambridge (Nobel Prize in 1984).
- Technique: Fuses an antibody-producing normal **B-lymphocyte** with an immortal cancerous **Myeloma cell** using polyethylene glycol (PEG) to form a hybrid cell called a **Hybridoma**.
- Cultured in HAT medium to produce limitless quantities of identical **Monoclonal Antibodies (mAbs)** directed against a single specific epitope; widely used in diagnostics (ELISA, pregnancy test kits) and targeted cancer immunotherapy.

- **Biotechnology Applications in Medicine**:
- **Recombinant Human Insulin (Humulin)**: First commercial recombinant biopharmaceutical product, produced by American pharmaceutical company **Eli Lilly in 1982**. Synthesised DNA sequences corresponding to human insulin A and B chains separately in *E. coli*, extracted them, and combined them via disulfide bonds. (Prior insulin extracted from slaughtered cattle and pigs caused severe allergic reactions).
- **First Clinical Gene Therapy (1990)**: Performed by **William French Anderson and Michael Blaese** on a 4-year-old girl (Ashanti DeSilva) suffering from **Severe Combined Immunodeficiency (SCID)** caused by **Adenosine Deaminase (ADA) deficiency**; retroviral vector used to introduce functional ADA cDNA into her cultured T-lymphocytes.
- **Recombinant Hepatitis B Vaccine**: Second-generation subunit vaccine produced by expressing Hepatitis B surface antigen (HBsAg) in the **yeast *Saccharomyces cerevisiae***.
- **Biotechnology Applications in Agriculture & Transgenic Crops**:
- **Bt Cotton**: Genetically engineered to resist the cotton bollworm (*Helicoverpa armigera*); carries ***cry1Ac*** and ***cry2Ab*** genes isolated from the soil bacterium ***Bacillus thuringiensis***; the bacterium produces crystalline inactive **Protoxin** proteins that solubilise in the **alkaline pH of the insect midgut**, binding to epithelial cells and creating perforations that cause larval death. Approved for commercial cultivation in India in **2002** by the **Genetic Engineering Appraisal Committee (GEAC)** under the MoEFCC.
- **Bt Corn**: Engineered with the ***cry1Ab*** gene to control the corn borer pest.
- **Golden Rice**: Genetically modified biofortified rice enriched with **β-carotene (precursor of Vitamin A)** to combat childhood blindness and Vitamin A deficiency (VAD); developed by **Ingo Potrykus and Peter Beyer** by introducing two genes (phytoene synthase from daffodil/maize and phytoene desaturase from the soil bacterium *Erwinia uredovora*).
- **Flavr Savr Tomato**: First commercially grown GM food crop (approved by US FDA in 1994); developed by Calgene using **Antisense RNA technology** to silence the gene for the enzyme **Polygalacturonase**, preventing cell wall pectin degradation and significantly delaying fruit softening/rotting.
- **RNA Interference (RNAi / Gene Silencing)**: Discovered by **Andrew Fire and Craig Mello (1998)** in *Caenorhabditis elegans* (Nobel Prize in 2006). Cellular defense mechanism using double-stranded RNA (dsRNA) to silence complementary mRNA transcripts post-transcriptionally via the Dicer-RISC pathway; used to produce transgenic tobacco plants resistant to the root-knot nematode ***Meloidogyne incognita***.
- **Transgenic Animals & Bioremediation**:
- **Dolly the Sheep (1996–2003)**: First mammal successfully cloned from an adult somatic body cell (mammary gland cell) via **Somatic Cell Nuclear Transfer (SCNT)** at the Roslin Institute in Edinburgh, led by **Sir Ian Wilmut** and Keith Campbell.
- **Transgenic Cow 'Rosie' (1997)**: Produced human protein-enriched milk containing **2.4 grams per litre** of human **α-lactalbumin**, nutritionally more balanced for human babies than natural cow milk.
- **Superbug (*Pseudomonas putida*)**: Genetically engineered multi-plasmid bacterium developed by Indian-American microbiologist **Dr. Ananda Mohan Chakrabarty (1971)**; capable of simultaneously degrading four major hydrocarbon components of crude oil (octane, hexane, xylene, naphthalene); milestone in bioremediation of marine oil tanker spills and first patented living organism (US Supreme Court, 1980).

- **Plasmids & Cloning Vectors**:
- **Plasmid Definition**: Small, circular, double-stranded, extrachromosomal DNA molecules found in bacteria (and occasionally archaea/eukaryotes) that replicate independently of chromosomal DNA.
- Plasmids often carry advantageous genes such as **antibiotic resistance** (e.g., ampicillin, tetracycline resistance).
- **Vector Function**: In genetic engineering, plasmids engineered to carry foreign DNA fragments into host cells are called **cloning vectors**. The hybrid molecule is termed **recombinant plasmid**.
- **Transformation**: The process by which foreign or recombinant DNA/plasmid is introduced into a recipient bacterial cell.
- **pBR322 Plasmid Vector**: Created in 1977 in the laboratory of **Herbert Boyer** at UCSF by researchers **Francisco Bolivar and Raymond Rodriguez**; named 'p' for plasmid, 'BR' for Bolivar & Rodriguez. Contains amp^R and tet^R marker genes and unique restriction cleavage sites.
- **Bacteriophages**: Bacterial viruses (e.g. bacteriophage λ, M13) used as vectors capable of accommodating larger foreign DNA inserts than standard plasmids.
- ***Agrobacterium tumefaciens***: Soil bacterium carrying the **Ti plasmid (Tumor-inducing)**; natural genetic vector extensively utilized for introducing foreign genes into dicotyledonous plant genomes.
- **Key Molecular Enzymes in Genetic Engineering**:
- **Restriction Endonucleases (Molecular Scissors / आणविक कैंची)**:
  - Cleave double-stranded DNA at or near specific palindromic sequences (restriction sites).
  - Discovered in *Haemophilus influenzae*; **Werner Arber, Hamilton O. Smith, and Daniel Nathans** were awarded the **1978 Nobel Prize in Physiology or Medicine**.
- **DNA Ligases (Molecular Stitchers / आणविक गोंद)**:
  - Facilitate the joining of DNA fragments by catalyzing the formation of covalent **phosphodiester bonds** between adjacent 3′-OH and 5′-phosphate ends.
- **In Vitro DNA Synthesis**: First synthesized *in vitro* by **Arthur Kornberg** in 1956 (awarded the **1959 Nobel Prize in Physiology or Medicine**).
- **Sequential Steps of Recombinant DNA (rDNA) Technology**: 1. *Step 1*: Identification and isolation of genetic material (DNA) from donor organism. 2. *Step 2*: Fragmentation of isolated DNA by specific restriction endonucleases. 3. *Step 3*: Isolation of desired DNA fragments and **Amplification of gene of**

interest using Polymerase Chain Reaction (PCR). 4. *Step 4*: Ligation of the desired DNA fragment into a suitable cloning vector (construction of recombinant DNA). 5. *Step 5*: Transfer / introduction of the recombinant vector into a competent host cell/organism (**Transformation**). 6. *Step 6*: Selection and screening of transformed recombinant host cells expressing the gene of interest. 7. *Step 7*: Culturing in bioreactors to obtain the foreign gene product at industrial scale, followed by **Downstream Processing** (purification, formulation, quality testing).

- **CRISPR-Cas9 Gene Editing**:
- **CRISPR** = *Clustered Regularly Interspaced Short Palindromic Repeats*; **Cas9** = *CRISPR-associated protein 9*, an RNA-guided DNA endonuclease acting as precision molecular scissors.
- Adapted from natural adaptive immune mechanisms in bacteria and archaea against bacteriophage viral infections.
- Employs a synthetic **guide RNA (gRNA)** that directs Cas9 to a target genomic locus, creating double-strand breaks for targeted gene knockout or knock-in.
- **Lulu and Nana Controversy (2018)**: Chinese biophysicist **He Jiankui** utilized CRISPR-Cas9 to edit the *CCR5* gene in human embryos to confer HIV resistance, resulting in the birth of twin girls (pseudonyms Lulu and Nana). This marked the world's first germline-edited babies, igniting global bioethical condemnation and moratoriums.
- **Stem Cells & Regenerative Medicine**:
- **Defining Properties**: (1) **Self-renewal** (ability to undergo continuous cell divisions while remaining undifferentiated); (2) **Potency** (capacity to differentiate into specialized cell types). Used to treat > 80 blood, bone, and immune disorders.
- **Types of Stem Cells**:
  1. **Embryonic Stem Cells (ESCs)**: Isolated from the **Inner Cell Mass (ICM) of blastocysts** (3–5 days post-fertilization, size of a dot). **Pluripotent** (can generate every cell type of the adult body, but cannot form placenta/umbilical cord). Bioethically controversial because derivation destroys the blastocyst.
  2. **Adult / Somatic Stem Cells**: Multipotent and tissue-specific; maintain and repair tissues in which they reside. E.g., **Hematopoietic stem cells (HSCs)** in bone marrow produce RBCs, WBCs, and platelets (do not form brain or liver cells). Autologous bone marrow harvesting is an ethical, non-controversial alternative to ESCs.
  3. **Induced Pluripotent Stem Cells (iPSCs)**: Adult somatic cells (e.g. skin fibroblasts) reprogrammed in vitro into embryonic-like pluripotent stem cells by introducing specific transcription factors (Oct3/4, Sox2, Klf4, c-Myc; pioneered by **Shinya Yamanaka**, Nobel Prize 2012).
  4. **Mesenchymal Stem Cells (MSCs / Stromal Cells)**: Multipotent stromal cells isolated from connective tissues (bone marrow, adipose fat, cord blood); differentiate into osteoblasts (bone), chondrocytes (cartilage), and adipocytes (fat).
  5. **Cord Blood Stem Cells**: Blood collected from placenta and umbilical cord post-delivery, rich in hematopoietic stem cells; non-invasive and ethically acceptable.
- **Amniocentesis (Amniotic Fluid Test / AFT)**:
- Prenatal diagnostic procedure typically performed at **14–16 weeks of gestation**.
- A syringe extracts a small sample of amniotic fluid containing shedding fetal cells from the amniotic sac.
- Used for karyotyping to diagnose chromosomal disorders (e.g., **Down Syndrome / Trisomy 21**), metabolic errors, and fatal fetal infections.
- Misused for prenatal fetal sex determination, leading to strict legal prohibition under India's **PCPNDT Act, 1994**.
- **DNA Fingerprinting (Genetic Profiling / DNA Typing)**:
- Developed in 1984 by British geneticist **Sir Alec Jeffreys**.
- **Scientific Basis**: **DNA Polymorphism** in non-coding repetitive DNA (**Variable Number Tandem Repeats / VNTRs / Minisatellites**), which show high individual-specific length variations inherited in a Mendelian fashion (50% from mother, 50% from father).
- First criminal conviction based on DNA fingerprinting: **Colin Pitchfork** (England, 1986).
- **Father of Indian DNA Fingerprinting**: **Dr. Lalji Singh** (established the Centre for DNA Fingerprinting and Diagnostics / CDFD at Hyderabad).
- Forensic biological samples: Blood, semen, saliva, hair follicles with root sheath, bone, teeth.
- Multi-coloured / glossy surface latent prints developed using **fluorescent organic powders**.
- **Animal Cloning Milestones & Somatic Cell Nuclear Transfer (SCNT)**:
- **Dolly the Sheep (1996–2003)**: Finn-Dorset ewe; world's first mammal cloned from an adult somatic body cell (mammary gland cell) via SCNT by **Ian Wilmut, Keith Campbell**, and team at the Roslin Institute, Scotland. Died in 2003 from progressive lung disease.
- **NDRI Karnal (National Dairy Research Institute, Haryana) Milestones**:
  - **Samrupa (6 Feb 2009)**: World's first cloned buffalo calf produced via hand-guided cloning (survived 6 days).
  - **Garima (6 Jun 2009)**: Female cloned buffalo calf, survived over 2 years (died August 2011).
  - **Garima-II (22 Aug 2010)**: Female cloned buffalo born from embryonic stem cells; later gave birth to normal healthy female calves **Mahima (2013)** and **Karishma (2014)**.
  - **Shrestha (26 Aug 2010)**: Male cloned buffalo calf from somatic cell of an elite bull.
  - **Sawarna (18 Mar 2013)**: Male cloned buffalo from somatic cell of semen.
  - **Deepasha (12 Dec 2014)**: India's first cloned endangered wild buffalo of Chhattisgarh (donor mother 'Asha' at Udanti Wildlife Sanctuary).
  - **Ganga (16 Mar 2023)**: India's first cloned indigenous **Gir breed** cow calf (birth weight 32 kg).
- **Injaz (2009)**: World's first cloned female camel, developed at Camel Reproduction Centre, Dubai, UAE, using ovarian cells.
- **Zhong Zhong & Hua Hua (2017)**: World's first cloned non-human primates (crab-eating macaques) produced via SCNT in China.
- **Genetically Modified Crops & Agrigenomics**:

- **Humulin (1982)**: First genetically engineered human hormone approved for consumer use (Eli Lilly / Genentech; recombinant human insulin synthesized in *E. coli*).
- **Bt Cotton**: Transgenic cotton containing insecticidal crystal protein genes from the soil bacterium ***Bacillus thuringiensis***.
  - *Bollgard I*: Single gene technology (**Cry1Ac**) approved in India in 2002 against American bollworm (*Heliothis armigera*).
  - *Bollgard II*: Double gene technology (**Cry1Ac + Cry2Ab**) approved in 2006, protecting against bollworms and *Spodoptera* caterpillars.
- **Bt Brinjal**: Transgenic brinjal containing the **Cry1Ac** gene developed by Mahyco against shoot and fruit borer (*Leucinodes orbonalis*); moratorium imposed by the Government of India in 2010 due to environmental and biosafety concerns.
- **GM Mustard (DMH-11 / Dhara Mustard Hybrid-11)**: Genetically modified hybrid *Brassica juncea* developed by Prof. Deepak Pental's team at the Center for Genetic Manipulation of Crop Plants (CGMCP), University of Delhi. Employs a three-gene system from soil bacterium ***Bacillus amyloliquefaciens***: **barnase** (male sterility), **barstar** (fertility restoration), and **bar** (glufosinate herbicide resistance), enabling commercial cross-pollination and hybridization.
- **Flavr Savr Tomato (1994)**: First commercially grown genetically engineered food granted US FDA approval; developed by Calgene. Contains an **antisense gene** that suppresses the endogenous fruit-softening enzyme **Polygalacturonase (PG)**, delaying rotting and prolonging shelf life while retaining vine-ripened flavor and color.
- **Golden Rice**: Genetically modified rice (*Oryza sativa*) engineered to biosynthesize **β-carotene (provitamin A)** in its edible endosperm to combat dietary Vitamin A deficiency and childhood blindness. Developed by **Prof. Ingo Potrykus and Dr. Peter Beyer** by inserting two foreign genes: ***psy*** **(phytoene synthase)** from daffodil (*Narcissus pseudonarcissus*) and ***crtI*** **(phytoene desaturase)** from the soil bacterium *Erwinia uredovora*.
- **Super Rice**: High-yielding rice developed by **Gurdev Singh Khush** at the International Rice Research Institute (IRRI), Los Baños, Philippines (awarded 1996 World Food Prize).
- **Terminator Seed Technology (v-GURT)**: Genetic Use Restriction Technology employing a tri-gene system that renders harvested seeds genetically sterile in subsequent generations, preventing farmers from saving and replanting seeds.
- **Microsatellite DNA (STRs)**: Short tandem repetitive non-coding DNA segments (1–6 bp) used as polymorphic genetic markers for studying evolutionary fauna relationships, paternity, and population genetics.
- **Transcriptome**: The complete repertoire of all messenger RNA (mRNA) transcripts expressed by the genome of an organism or tissue at a given moment.
- **Pleiotropism**: Phenomenon wherein a single gene simultaneously influences two or more distinct phenotypic traits (e.g. **Sickle Cell Anaemia**, Phenylketonuria, Marfan Syndrome).

Bilingual Terminology -- Cell, Genetics and Biotechnology

English Term	Hindi Term	Key Anchor
Cell (कोशिका)	कोशिका	Basic structural and functional unit of life
Prokaryotic Cell (प्रोकैरियोटिक कोशिका)	आदिकोशिकीय	No membrane-bound nucleus; e.g., bacteria, cyanobacteria
Eukaryotic Cell (यूकैरियोटिक कोशिका)	सुकोशिकीय	Membrane-bound nucleus; plants, animals, fungi, protists
Cell Wall (कोशिका भित्ति)	कोशिका भित्ति	Cellulose (plants), Chitin (fungi), Peptidoglycan (bacteria)
Cell Membrane (कोशिका झिल्ली)	कोशिका झिल्ली	Fluid mosaic model; semi-permeable phospholipid bilayer
Nucleus (केन्द्रक)	केन्द्रक / नाभिक	Control centre of cell; contains chromosomes and DNA
Mitochondria (माइटोकॉन्ड्रिया)	माइटोकॉन्ड्रिया	"Powerhouse of cell"; ATP synthesis via cellular respiration
Chloroplast (हरितलवक)	हरितलवक / क्लोरोप्लास्ट	Site of photosynthesis; contains chlorophyll; has own circular DNA
Ribosome (राइबोसोम)	राइबोसोम	Site of protein synthesis; 70S (prokaryote), 80S (eukaryote)
Endoplasmic Reticulum (अन्तःप्रद्रव्यी जालिका)	अन्तःप्रद्रव्यी जालिका	Rough ER = protein transport; Smooth ER = lipid synthesis
Golgi Body (गॉल्जी काय)	गॉल्जी काय	"Post Office of cell"; processes and packages proteins/lipids
Lysosome (लाइसोसोम)	लाइसोसोम	"Suicide bag"; contains digestive enzymes; autophagy
Vacuole (रिक्तिका)	रिक्तिका	Storage organelle; large central vacuole in plant cells
Chromosome (गुणसूत्र)	गुणसूत्र	Thread-like structures carrying genes; DNA + histone proteins
Gene (जीन)	जीन	Unit of heredity; segment of DNA encoding a protein
DNA (डीऑक्सीराइबोन्यूक्लिक अम्ल)	डीऑक्सीराइबोन्यूक्लिक अम्ल	Double helix; deoxyribose sugar; bases ATGC
RNA (राइबोन्यूक्लिक अम्ल)	राइबोन्यूक्लिक अम्ल	Single stranded; ribose sugar; uracil replaces thymine
Mitosis (समसूत्री विभाजन)	समसूत्री विभाजन	Cell division for growth; 2 identical daughter cells; PMAT
Meiosis (अर्धसूत्री विभाजन)	अर्धसूत्री विभाजन	Reduction division; 4 haploid cells; genetic recombination
Mutation (उत्परिवर्तन)	उत्परिवर्तन	Permanent change in DNA sequence; basis of evolution
Genetic Engineering (आनुवंशिक अभियांत्रिकी)	आनुवंशिक अभियांत्रिकी	Deliberate modification of organism's genome
Recombinant DNA (पुनर्संयोजित डीएनए)	पुनर्संयोजित डीएनए	DNA formed by combining DNA from different sources
Biotechnology (जैव-प्रौद्योगिकी)	जैव-प्रौद्योगिकी	Use of living organisms/systems for products and services

English Term	Hindi Term	Key Anchor
Plasmid (प्लाज्मिड)	प्लाज्मिड	Small circular DNA in bacteria; used as cloning vector
Restriction Enzyme (प्रतिबंध एंजाइम)	प्रतिबंध एंजाइम	"Molecular scissors"; cuts DNA at specific sequences
PCR (पॉलीमरेज श्रृंखला अभिक्रिया)	पॉलीमरेज श्रृंखला अभिक्रिया	Amplifies specific DNA segments; invented by Kary Mullis (1983)
CRISPR-Cas9 (क्रिस्पर-कैस9)	जीनोम संपादन तकनीक	Genome editing tool; 2020 Nobel Chemistry (Charpentier & Doudna)
Clone (क्लोन)	क्लोन	Genetically identical organism produced asexually
Transgenic Organism (ट्रांसजेनिक जीव)	पारजीनी जीव	Organism with foreign gene inserted into its genome
Bt Crop (बीटी फसल)	बीटी फसल	Carries Bacillus thuringiensis gene for insect resistance
Stem Cell (मूल कोशिका)	मूल कोशिका / स्टेम सेल	Undifferentiated cell that can become any cell type
Genome (जीनोम)	जीनोम	Complete set of genetic information of an organism
Human Genome Project (मानव जीनोम परियोजना)	मानव जीनोम परियोजना	Completed 2003; mapped all ~3 billion base pairs of human DNA
Bioremediation (जैव-निराकरण)	जैव-उपचारण	Use of microorganisms to clean up pollution

One-Liner Revision -- Cell, Genetics and Biotechnology

#	Fast Fact
1	Cell theory = Schleiden (1838) + Schwann (1839) + Virchow (1855)
2	Smallest cell = Mycoplasma (0.1-0.3 micron); Largest single cell = Ostrich egg
3	Prokaryote = **NO nucleus**, NO membrane-bound organelles; e.g., bacteria
4	Eukaryote = **TRUE nucleus** + membrane-bound organelles; e.g., plants, animals
5	Mitochondria = "Powerhouse of cell" (ATP synthesis)
6	Chloroplast = "Kitchen of cell" (photosynthesis); has own DNA + 70S ribosomes
7	Ribosome: Prokaryote = **70S** (50S+30S); Eukaryote = **80S** (60S+40S)
8	Golgi body = "Post Office / Traffic police" of cell (protein packaging and dispatch)
9	Lysosome = "Suicide bag" (autolysis); contains hydrolytic enzymes
10	DNA bases: **A-T** (adenine-thymine) and **G-C** (guanine-cytosine)
11	RNA has **Uracil** instead of Thymine; ribose sugar instead of deoxyribose
12	DNA double helix discovered by **Watson and Crick** (1953); X-ray by Rosalind Franklin
13	Human chromosomes = **46** (23 pairs); sex chromosomes = XY (male), XX (female)
14	Mitosis = 2 identical daughter cells (diploid); for growth and repair
15	Meiosis = 4 haploid daughter cells; for gamete formation; crossing over in Prophase I
16	Mutation = permanent change in DNA; caused by radiation, chemicals, viruses
17	Restriction enzymes = "Molecular scissors"; EcoRI is most famous (cuts at GAATTC)
18	PCR invented by **Kary Mullis (1983)**; Nobel Chemistry 1993
19	First recombinant DNA = Cohen and Boyer (1973); using EcoRI
20	First transgenic plant = tobacco (1983); first transgenic animal = mouse
21	Dolly the sheep = first cloned mammal from adult somatic cell (Wilmut, 1996, UK)
22	Bt gene from **Bacillus thuringiensis**; Bt crops resist Lepidoptera/Coleoptera pests
23	Human Genome Project completed = **April 2003** (~3.2 billion base pairs, ~20,000-25,000 genes)
24	CRISPR-Cas9 = Nobel Chemistry **2020** (Emmanuelle Charpentier + Jennifer Doudna)
25	Golden Rice = transgenic rice with beta-carotene gene; developed to address Vitamin A deficiency
26	Insulin produced by GM bacteria = **Humulin** (Eli Lilly, 1982); first biotech drug
27	Chakrabarty's superbug (1980) = 1st GMO to get US patent; *Pseudomonas putida*
28	RNA interference (RNAi) = Nobel 2006 (Fire and Mello); gene silencing mechanism
29	Stem cells: Totipotent (zygote) > Pluripotent (embryo) > Multipotent (adult)
30	Bioremediation = microorganisms clean pollutants; Chakrabarty's *Pseudomonas* = oil spill